

PCR-5D5

Symmetry and Charge Extensions of the Neutrino Standing-Carrier Space

Exact $B - L$ Fibre Stratification, Symmetry-Bearer Structure, and Target-Indexed Comparison with Active Closure

Let $\mathcal{F}_{D5}$ be the fixed seven-member PCR-5D5 target family, and let $\rho_\tau : Q_{D5}(\tau) \rightarrow Q_{osc}^{SCIC}$ be the restriction map for target τ . At the symbolic source scope inherited from PCR-3 and PCR-5B, each target admits a complete source representation, a simultaneous one-apex physical realization criterion, a retained-oscillation restriction, and an exact groupoid-valued lift atlas. The target-indexed base images are

$$\begin{aligned}
\text{im } \rho_{acc}^{(4)} &= Q_{Acc \leq 4}, & \text{im } \rho_{bg, min} &= Q_{MinBG}, \\
\text{im } \rho_{loc, 3N} &= Q_{Loc_{3N}}, & \text{im } \rho_{glob} &= Q_{GlobProt}, \\
\text{im } \rho_{exp[\mathcal{O}]} &= Q_{ExpLift(\mathcal{O})}, & \text{im } \rho_{sp[q_\Phi]} &= Q_{Cov_\Phi(q_\Phi)}, \\
\text{im } \rho_{rem[\Gamma, q_\Phi]} &= Q_{RemFaith(\Gamma, q_\Phi)}.
\end{aligned}$$

The first four images are nonempty proper-continuum subatlases. For the local and exact-global images, the accepted-side continuum is completed through an explicit source-formed finite-order RG/EFT matching right inverse that fixes the entire target-retained light continuation and leaves only the physical threshold visible in M^{match} . The explicit-breaking image consists exactly of the target-bearing lifts in $Q_{ExpLift(\mathcal{O})}$: a physically valid $B - L$ -violating spectator with no neutrino-matching incidence is removed by activity reduction and cannot furnish the required target record. A universal neutrino-matched, oscillation-transparent full-base branch would be sufficient for full-base explicit breaking, but no such universal theorem is established here. For the explicit, spontaneous, and remnant targets, cardinality remains NotReady wherever no separate lower-bound theorem is available.

If an accepted D5 point carries an exact continuous or exact residual action on the complete target-relevant mass-response and spectral-incidence structure, then charge or residual character is constant on each connected spectral component. This is *connected spectral charge uniformity*; it is a symmetry-bearer result, not a theorem of terminal-carrier identity. Bounded accidental conservation does not imply this conclusion when the complete mass response violates $B - L$, and terminal promotion remains NotActivated.

The D3–D5 comparison is likewise target-indexed. At THUCL-D3-1.3, accidental and exact-global $B - L$ standing are IndependentCoordinate relative to the active-closure separator. The minimal-background, local-3N, explicit-breaking, spontaneous, and remnant comparisons remain NotReady. Any common or emergent generator must therefore be established, if at all, by the simultaneous PCR-5E target.

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26 August 2026

Abstract

PCR-5B identifies the standing-carrier quotient of the fixed neutrino-oscillation target as an exact mixed moduli object with at least continuum many SCIC classes, and PCR-5C supplies the target-extension and lift-fibre calculus. The completed D1–D4 extensions show that absolute kinematic scale, lepton-number-violating response, active-sector closure, and cosmological transport act on distinct standing coordinates. This paper determines the final marginal extension before the joint neutrino target: symmetry, charge, anomaly, breaking, and remnant $B - L$ standing.

The analysis distinguishes seven non-equivalent targets,

$$\mathcal{F}_{D5} = \{T_{\text{acc}}^{(4)}, T_{\text{bg,min}}, T_{\text{loc},3N}, T_{\text{glob}}, T_{\text{exp}}[\mathcal{O}], T_{\text{sp}}[q_\Phi], T_{\text{rem}}[\Gamma, q_\Phi]\}.$$

The comparison class is the B15-framed PCR-3 carrier theory conservatively extended by SOE–D5–1. For every target and B15 frame, the analysis determines the source groupoid, the simultaneous physical realization category, the retained-oscillation restriction, the connected homotopy lift groupoids, lawful charge and anomaly descent, and the breaking and remnant quotients. Failures remain in typed complements and are not reinterpreted as zeros, inapplicability, or different target fibres.

The exact base images are source-semantic subatlases. $Q_{\text{Acc} \leq 4}$ consists of classes whose complete active $d \leq 4$ operator content preserves $B - L$; Q_{MinBG} consists of classes carrying the minimal chiral background-anomaly packet; $Q_{\text{Loc},3N}$ consists of classes admitting a faithful unbroken local $U(1)_{B-L}$ action with one active right-handed neutrino per family, covariant support, and complete local anomaly cancellation; and Q_{GlobProt} consists of classes admitting a faithful global action with a continuation-stable protector. Each of these four images is nonempty and proper and contains continuum subatlases on both the accepted and excluded sides. The accepted-side continuum for the local and exact-global targets is realized by an exact physical-response vectorlike Dirac family together with an explicit source-formed finite-order RG/EFT matching right inverse: all target-retained light continuations are common, while the physical heavy threshold remains visible in the oriented matching coordinate. The local and exact-global targets exclude active Weinberg and renormalizable Majorana source support, whereas the bounded accidental target admits the Weinberg EFT while excluding renormalizable $\Delta L = 2$ terms.

For explicit breaking, the exact image consists of classes admitting a source-active operator or process whose D5 matching incidence lies on an actual neutrino carrier, mass-response, or matching path and whose physical realization leaves the retained oscillation packet unchanged. A physically valid $B - L$ -violating sector without this target-bearing neutrino path is a spectator and does not furnish a D5 lift. Intrinsic target-bearing examples, including the Weinberg source, establish nonvacuity, but no universal neutrino-matched transparent full-base theorem is presently available; generic explicit-breaking cardinality therefore remains NotReady. Spontaneous breaking by an integral charge q_Φ is classified by the condition that every active charge violation lie in the ideal generated by q_Φ , together with the independent action, anomaly/global, vacuum, stabilizer, matching, and continuation equations. For several breaking charges, the faithful remnant is obtained from the charge-lattice quotient or Smith normal form, and retained operators survive exactly when their violations vanish in that quotient. No universal singleton exclusion is claimed for explicit, spontaneous, or remnant targets without a separate cardinality theorem.

The bearer theorem addresses the carrier/spectral distinction exposed by D1–D4. Let $M_\nu = \sum_i m_i P_i$ be a standing spectral resolution, and let the charged-current boundary and source/-matching incidences define a graph on the P_i . Whenever the target supplies exact continuous

or exact residual covariance of the complete mass-response and spectral-incidence structure, charge or residual character is constant on every connected component. This *connected spectral charge uniformity* is compatible with a common continuation-bearing carrier but does not establish terminal-carrier identity. The bounded accidental $d \leq 4$ target does not inherit the result when the complete mass response violates $B - L$.

The comparison with the PCR-5D3 successor target is also target-indexed. Explicit deformation families prove marginal independence for the accidental and exact-global targets. The minimal-background, local- $3N$, explicit-breaking, spontaneous, and remnant comparisons remain NotReady, and no common-generator factorization follows from marginal image relations. The resulting target-tagged fibres, cardinality conclusions, separators, bearer classifications, source-composition profiles, failure frontiers, and D3 comparison data define the D5 marginal input to the simultaneous PCR-5E analysis.

The result is an exact target-relative classification at the finite source-presentation and symbolic B15 scope. It does not assert that any $B - L$ target is realized in nature, derive a numerical gauge coupling or breaking scale, infer three terminal neutrino carriers from three masses, exclude singletonhood where no cardinality theorem has been proved, or complete the separately defined PCR-5D3 THUCL-D3-1.1 programme. Supplementary material documents provenance, exact arithmetic, and reproducibility; the scientific claims rest on the definitions and proofs presented here.

Keywords. AASC; neutrino oscillation; standing carrier; $B - L$; anomaly cancellation; target extension; remnant symmetry; spectral-sector standing.

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1 Problem, logical position, and scope

1.1 The remaining marginal question

The neutrino target-extension sequence begins with a fixed oscillation target and an exact standing-carrier quotient. That quotient does not count fields, poles, mass labels, source families, or ultraviolet mechanisms. It identifies complete target-relative carrier standing through the retained seven-channel packet and its licensed continuations. An additional phenomenon may eliminate a base class, preserve it, split it by a newly standing coordinate, or require a source relation that was invisible at the oscillation target.

PCR-5D5 addresses the final marginal question before the joint target:

When $B - L$, charge, anomaly, breaking, and remnant information acquire standing, which oscillation-carrier classes admit lifts, how are those fibres stratified, and what physical structure bears the new standing?

The problem is therefore not whether a familiar model can be assigned a $B - L$ label. It is the exact transformation of the already classified oscillation carrier space under a target-complete symmetry and charge extension.

1.2 Dependency order

The logical sequence is

$$\text{PCR} - 5\text{B} \rightarrow \text{PCR} - 5\text{C} \rightarrow \text{PCR} - 5\text{D1} - \text{D5} \rightarrow \text{PCR} - 5\text{E} \rightarrow \text{PCR} - 6.$$

PCR-5B provides the exact B15-framed base atlas, and PCR-5C establishes restriction, no merger, connected-fibre reconstruction, composition, source composition, joint-target formation, and failure provenance [4, 5]. D1–D4 provide target-qualified comparisons but do not determine D5 membership [6, 7, 8, 9]. Their synthesis motivates the hypothesis of one continuation-bearing neutrino carrier with several standing spectral sectors, while expressly withholding that hypothesis from use as a D5 selector. The D5 images and bearer classifications must therefore be derived from the D5 source and target structures themselves.

No conclusion from D1, D2, D3, D4, the Standard Model carrier closure, the generation-cardinality theorem, or the Standard Model symmetry atlas determines membership in a D5 lift fibre unless that datum belongs to the fixed D5 target and is realized on the same enriched physical point.

Proof. A target extension retains the base records and adds target-typed records. D1, D2, D3, and D4 have different added records and different acceptance maps. Equal or nested marginal images do not identify those maps. The Standard Model carrier and symmetry results classify their own fixed targets and cannot supply D5 source activity, physical co-instantiation, or target-specific protection. Any attempted use of one of those conclusions as a D5 membership rule is either an undeclared target coordinate, a second classifier, or a target update. PCR-5C no-backflow and fixed-domain standing therefore prohibit the move. $\square$

1.3 Three multiplicities kept separate

Three distinct cardinalities occur throughout the analysis:

1. the number of forced target roles;
2. the cardinality of the global standing-carrier quotient $Q_{\text{osc}}^{\text{SCIC}}$;
3. the number of standing spectral sectors within one realized carrier.

They are not interchangeable. In particular,

$$|\text{Spec}(M_\nu)| = 3 \not\Rightarrow |Q_{\text{terminal carrier}}| = 3.$$

A D5 target may act uniformly on a carrier, distinguish spectral subobjects, or be borne by a relation among several source sectors. Only a later terminal-identity and Role-Occupancy Closure analysis can promote a subobject to a terminal carrier.

1.4 Structure of the argument

The proof has five stages.

1. **Target formation.** The seven symmetry and charge questions are defined as pairwise non-equivalent target objects.
2. **Source and physical realization.** The SOE–D5–1 source class is represented over every B15 frame and realized on a single semantic physical apex.
3. **Retained-base descent.** The oscillation packet is evaluated with the D5 structure present, and the restriction map descends to SCIC.
4. **Target-specific classification.** Charge, anomaly, local/global, breaking, remnant, fibre, image, and overlap questions are resolved symbolically.
5. **Marginal interpretation.** Bearer standing and the D3 comparison are established without terminal promotion, and the resulting D5 datum is prepared for the simultaneous PCR-5E target.

The order is essential. A base-image claim made before simultaneous physical realization and retained-oscillation transparency would count source descriptions rather than lawful target lifts.

1.5 Main theorem

At the target/source scope fixed below:

1. every fixed D5 source point belongs to an exact source and physical groupoid or a typed complement;
2. retained oscillation restriction is well defined and no two distinct base SCIC classes merge;
3. the seven target images are the exact subatlases displayed in the Principal Result block;
4. charge and anomaly records descend through lawful lattice, basis, family, and presentation transformations;

5. explicit, spontaneous, and remnant breaking are classified by source activity, vacuum stabilizers, charge-lattice ideals, and faithful quotients;
6. D5 bearer status is total and never itself yields a terminal carrier;
7. D3–D5 comparison authority is target-indexed: accidental and exact-global standing are marginally independent of the D3-1.3 active-closure separator, whereas the minimal-background, local-3N, explicit-breaking, spontaneous, and remnant rows remain NotReady;
8. the complete target-tagged D5 marginal object is supplied to PCR-5E with all unresolved authority boundaries retained.

The remainder of the paper proves these clauses in dependency order.

1.6 Exact nonclaims

Nonclaim 1.1. The analysis does not prove that $B - L$ is an exact law of nature, that it is gauged, that a particular breaking scale is realized, or that a particular remnant survives cosmological history.

Nonclaim 1.2. The local three- ν_R target requires one active singlet per family by target definition. The number three is not inferred from generation governance alone and is not a primitive carrier count.

Nonclaim 1.3. A null $0\nu\beta\beta$ search, a vanishing parameter value, or a missing event is not an exact global protector.

Nonclaim 1.4. The D3 comparison uses only the target-qualified successor result at THUCL-D3-1.3. It neither completes nor updates the separately defined THUCL-D3-1.1 programme.

2 Foundational setting and target identity

2.1 The exact oscillation base

The base object is the groupoid-valued PCR-5B atlas [4]

$$\mathcal{A}_{\text{osc}}^{\text{exec}} \simeq \mathcal{N}_{5,\text{osc}}^{\text{exec}}, \quad \pi_0 \mathcal{A}_{\text{osc}}^{\text{exec}} \cong Q_{\text{osc}}^{\text{SCIC}},$$

with complete retained packet

$$\Xi_{\text{osc}} = (T^{\text{face}}, R^{\text{stand}}, \Delta_{\text{fr}}^{\text{del}}, C^{\text{act}}, M^{\text{match}}, B^{\text{bdy}}, F^{\text{same}}).$$

For each source presentation $a = (F, \sigma)$, the B15 frame records the raw source domain, source equations, applicability, total status, the PCR-4 survivor object, lawful presentation groupoid, standing-core functor, packet moduli, provenance, fixed no-repair deletion, auxiliary source-composition data, typed failure frontier, separators, and all pairwise and higher overlaps. D5 is evaluated over these complete frames rather than over a reduced list of representative models.

Definition 2.1 (Base representative category). For $q \in Q_{\text{osc}}^{\text{SCIC}}$, let $\mathcal{N}_{\text{osc}}[q]$ be the connected full subgroupoid of concrete standing cores representing q . A D5 lift fibre is formed over this connected groupoid, not over one preferred chart or source label.

2.2 Imported theorem roles

Source	Permitted imported role	Forbidden use
PCR-3	Finite source-first carrier ontology, source coherence, activity reduction, semantic normal forms.	Candidate completeness is not D5 survival.
PCR-4	One-point simultaneous physicality and common-cone discipline.	Marginal records may not be spliced.
PCR-5B/B15	Exact base atlas, groupoids, packet, overlaps, complements, continuum structure.	D5 may not alter base identity.
PCR-5C	Restriction, no merger, connected fibres, homotopy reconstruction, source composition, joint targets.	It supplies no D5-specific physics.
D1–D4 synthesis	Target-qualified standing-coordinate comparison and carrier/spectral firewall.	The one-carrier hypothesis is not a membership rule.
SYM-7	Four neutrino source targets; accidental, Majorana, Weinberg, local anomaly, and local/global frontier theorems.	Local cancellation is not global completion.
SYM-8	Vacuum-orbit, stabilizer, and residual-action discipline.	A chosen representative or vev is not a physical branch.
SYM-9	Global-form, line/defect, background, and family-indexed completion packet.	Inactive extended records cannot be imported.
SYM-11	Source-first symmetry normal forms and closure at the Standard Model target.	It does not replace the D5 extension.
SM-3–SM-6	Neutral weak matter, charge, flavour, Higgs-bridge, and generation-governance roles.	Mass labels and generations do not self-promote to carriers.
SCIC and ROC	Same-scope carrier identity and terminal compatible-occupancy discipline.	Neither authorizes terminal promotion within D5.

2.3 Target-specificity and dependency control

Every imported statement retains its exact target tag. In particular, none of the following implications is valid:

$$\begin{aligned}
 &\text{mixed-gauge cancellation} \Rightarrow \text{cubic/gravitational cancellation,} \\
 &\text{local polynomial cancellation} \Rightarrow \text{global completion,} \\
 &\text{accidental conservation through } d \leq 4 \Rightarrow \text{exact global protection,} \\
 &\text{nonzero breaking field} \Rightarrow \text{faithful remnant,} \\
 &N_g = 3 \Rightarrow 3\nu_R \text{ occupancy,} \\
 &\text{im } \rho_{D3} = \text{im } \rho_{D5} \Rightarrow \text{one common generator.}
 \end{aligned}$$

Adding a charge lattice, global quotient, anomaly probe, breaking carrier, operator-acceptance rule, or continuation absent from the declared target is a `TargetUpdate`. Introducing a new source sort or constitutive relation is a `SourceUpdate`.

2.4 Status calculus

Every target/source point has exactly one status in

$$\{\text{Pass, Fail, N/A, Undef, ROOS, SourceUpdate, TargetUpdate}\}.$$

N/A is certified from target typing before evaluation. ROOS denotes a lawful source object outside the response regime of the target. Undef is a fail-closed active evaluation failure. Source and target updates are changes of scope, not negative candidates, and do not enter the survivor quotient.

2.5 Mathematical authority and supplementary checks

The scientific claims are established by the definitions and proofs in the paper. Supplementary materials reproduce the anomaly arithmetic, identify the exact target and dependency versions used, and check consistency among the marginal conclusions. They document provenance and reproducibility but have no independent authority for source completeness, physical existence, fibre exhaustion, or any base-image theorem.

3 Seven symmetry and charge targets

3.1 Why $B - L$ does not define a single target

Following the target-separation results of SYM-7–SYM-9 [12, 13, 14], the phrase “ $B - L$ symmetry” may refer to several non-equivalent physical claims: an accidental selection rule in a bounded operator envelope, a background anomaly diagnosis, an unbroken local gauge action, an exact global protector, explicit violation, spontaneous breaking, or a remnant after breaking. These claims require different records, have different failure loci, and descend through different lawful quotients. They must therefore be treated as distinct target fibres.

Definition 3.1 (D5 target index).

$$\mathcal{F}_{\text{D5}} = \{T_{\text{acc}}^{(4)}, T_{\text{bg,min}}, T_{\text{loc,3N}}, T_{\text{glob}}, T_{\text{exp}}[\mathcal{O}], T_{\text{sp}}[q_{\Phi}], T_{\text{rem}}[\Gamma, q_{\Phi}]\}.$$

The target tag is part of the object identity. Equality of two later base images does not identify their targets.

3.2 Complete target records

For $\tau \in \mathcal{F}_{\text{D5}}$, define

$$\mathbb{T}_{\tau} = (T_{\text{osc}}, \tau, \mathcal{A}_{\tau}, \Lambda_{\tau}, \mathcal{O}_{\tau}, \mathcal{P}_{\tau}, \mathcal{G}_{\tau}, \mathcal{V}_{\tau}, \mathcal{M}_{\tau}, \mathcal{B}_{\tau}, \text{Acc}_{\tau}, \partial\text{Fail}_{\tau}).$$

Here $\mathcal{A}_{\tau}$ is the symmetry-action and bearer packet; Λ_{τ} the faithful charge lattice and normalization class; $\mathcal{O}_{\tau}$ the target-active operator packet; $\mathcal{P}_{\tau}$ the local or background obstruction packet; $\mathcal{G}_{\tau}$ the global-form and extended packet; $\mathcal{V}_{\tau}$ the vacuum, breaking, stabilizer, and remnant packet; $\mathcal{M}_{\tau}$ the matching, threshold, and continuation packet; $\mathcal{B}_{\tau}$ the bearer-resolution interface; and $\partial\text{Fail}_{\tau}$ the typed failure and update frontier.

3.3 Target definitions

Definition 3.2 (Renormalizable accidental target). $T_{\text{acc}}^{(4)}$ asks whether every source-active operator of canonical dimension $d \leq 4$ in the declared D5 operator envelope is invariant under the fixed $B - L$ charge action. Higher-dimensional operators remain recorded but do not alter the bounded $d \leq 4$ statement.

Definition 3.3 (Minimal background target). $T_{\text{bg},\text{min}}$ fixes the minimal Standard Model chiral inventory, the left-handed Weyl convention, a background $B - L$ probe, and the complete six-row local/background anomaly packet. A change in the chiral inventory defines a different target.

Definition 3.4 (Local three-singlet target). $T_{\text{loc},3N}$ fixes a faithful unbroken local $U(1)_{B-L}$ action, one chiral right-handed neutrino per family in the anomaly-cancelling packet, complete covariant target-active support, and vanishing applicable local polynomial anomalies. Additional anomaly-neutral vectorlike support is permitted when it preserves the faithful action, the anomaly packet, and the retained neutrino matching. The target does not fix a global group, a breaking sector, or a line/defect packet.

Definition 3.5 (Exact global protector target). T_{glob} requires a faithful global $B - L$ action on every target-active field and operator, together with a continuation-stable protector relation excluding every target-active violation route. A written charge label, a zero coefficient, or finite non-observation is not a protector.

Definition 3.6 (Explicit-breaking target). $T_{\text{exp}}[\mathcal{O}]$ is indexed by a source-admitted operator or process class $\mathcal{O}$. It requires non-null physical activity, nonzero $B - L$ violation, target-bearing neutrino matching, and licensed continuation. An available but inactive operator lies in the typed complement.

Definition 3.7 (Spontaneous-breaking target). $T_{\text{sp}}[q_\Phi]$ fixes a breaking carrier Φ of faithful integral charge q_Φ , a physical vacuum branch with its orbit and stabilizer, and covariant matching of every target-active operator. Explicit breaking not absorbed into an invariant pre-breaking source is a failure or belongs to a separate target.

Definition 3.8 (Remnant target). $T_{\text{rem}}[\Gamma, q_\Phi]$ extends the spontaneous target by a faithful residual action Γ , the charge-lattice quotient, operator selection rules, and any activated global-form, line, defect, or background records. The remnant is derived from the breaking data; it is not an independent label.

3.4 Completeness and separation

Theorem 3.9 (D5-01 – Target-family completeness). *Within the fixed D5 comparison scope, every target-admitted symmetry/charge question belongs to exactly one of the seven target fibres or requires an explicit target update. Every lawful target morphism preserves the tag, action/bearer packet, anomaly probe, operator envelope, global-form branch, breaking data, remnant quotient, and acceptance rule.*

Proof. The target specification partitions the questions according to the physical record whose standing is at issue: bounded-envelope invariance, background obstruction, local gaugeability, exact global protection, explicit violation, spontaneous breaking, or residual symmetry. These predicates are pairwise non-equivalent because each pair differs in at least one required record or acceptance condition. A lawful target morphism must preserve every retained record; changing one changes the target fingerprint. The seven cases exhaust the declared target grammar. $\square$

Corollary 3.10 (No cross-fibre backflow). *No theorem proved in one D5 fibre can be transported to another by dropping or relabeling target fields. In particular, local cancellation does not prove exact global protection, accidental conservation does not prove an exact symmetry, and a remnant does not prove the original continuous action remains unbroken.*

3.5 Target overlaps

Two target fibres may contain the same physical realization only when that realization carries both complete target packets. The result is a multi-target object, not an identification of target tags. For example, a Dirac three-singlet realization may instantiate both $T_{\text{acc}}^{(4)}$ and T_{glob} , while a spontaneously broken realization may also instantiate a remnant target. The overlap is represented by the fibre product

$$\mathcal{N}_{D5}(\tau_1 \wedge \tau_2) \simeq \mathcal{N}_{D5}(\tau_1) \times_{\text{Phys}_{D5}}^h \mathcal{N}_{D5}(\tau_2).$$

Such an object belongs to an explicitly declared multi-target comparison or to PCR-5E; it does not merge the marginal fibres.

4 Source category and candidate atlas

4.1 Semantic comparison class

Fix a B15-framed base source $H \in \text{Src}_{D5}^{\text{base}}$ and a target $\tau \in \mathcal{F}_{D5}$. A finite semantic D5 source realization over H is

$$X = (H; \mathcal{S}, \mathcal{Q}, \mathcal{A}, \mathcal{O}, \mathcal{G}, \mathcal{V}, \mathcal{C}, \mathcal{B}, \text{Prov}),$$

where the added entries are, respectively, symmetry actions, charge objects, anomaly records, operator support, global form, vacuum/breaking/remnant data, protector/violation and matching/continuation relations, bearer incidence, and provenance. The comparison class is specified by this physical source content before a formal normal form is chosen.

The scope includes every finite same-target source architecture whose active D5 records belong to these categories. An essentially infinitary proposal must admit a finite symbolic presentation to enter the comparison class; otherwise it receives SourceUpdate rather than being silently excluded.

4.2 The conservative source theory

The theory SOE–D5–1 retains every PCR-3/B15 source sort [1, 4] and adds the dependent sorts

SymAction, ChargeObj, AnomalyRecord, GlobalForm, Break, Stabilizer,
Remnant, Protect, Violate, BearerIncidence.

Its constitutive equations express action and charge covariance, complete applicable anomaly sums, operator charge balance, vacuum stabilizer conditions, remnant quotients, matching/RG/threshold transport, global-form and extended-sector compatibility, and higher-arity bearer incidence.

Definition 4.1 (Exact source reduct). The functor

$$U_{D5} : \text{SOE–D5–1} \longrightarrow \text{Src}_{D5}^{\text{base}}$$

forgets D5-only cells and returns the original B15 source object. It preserves every base

node, relation, source equation, higher-arity incidence, boundary, matching arrow, and provenance record.

Definition 4.2 (Canonical source inclusion). The functor

$$I_{D5} : \text{Src}_{D5}^{\text{base}} \longrightarrow \text{SOE-D5-1}$$

sends a base source H to the same B15 object equipped with the target-typed null D5 data and no new source-active D5 incidence. It is the identity on all base nodes, relations, equations, higher-arity incidences, boundaries, matching arrows, and provenance records. Consequently,

$$U_{D5} \circ I_{D5} = \text{id}_{\text{Src}_{D5}^{\text{base}}}.$$

4.3 Activity reduction

D5 activity is evaluated pointwise at source level. A D5 incidence is removed only when its source-defined activity is exactly null; every non-null target-active incidence is retained without a magnitude threshold. This reduction precedes bearer support, target-path closure, and source equivalence.

The rule prevents two opposite errors. A vanishing operator coefficient does not become a physical explicit-breaking source, while a small nonzero breaking or gauge coupling is not removed merely because it is negligible at a different target.

4.4 Neutrino-matched, oscillation-transparent explicit breaking

The fixed explicit-breaking target requires more than a physically valid $B - L$ -violating sector. Its matching record must place the violation on a target-bearing neutrino path, while retained-oscillation transparency requires that the same D5 incidence have zero output under every retained evaluator and licensed continuation. These are independent conditions.

Definition 4.3 (Neutrino-matched transparent predicate). For a B15 source frame H and an explicit-breaking operator or process type $\bar{O}$, write

$$\text{NMTr}_{D5}(\bar{O}; H) = 1$$

when there exists a finite enriched source point obtained from H by adjoining a D5 block $B_{\bar{O}, H}$ and a source-active matching/bearer incidence

$$\text{Match}_{\nu}^{D5}(B_{\bar{O}, H}, H) \neq \emptyset$$

such that:

1. the exact source reduct is H , and no base node, relation, matching arrow, or standing coordinate is changed;
2. $\bar{O}$ is source-active, has $\delta_{\bar{O}} \neq 0$, and carries the fixed operator identity, source support, activity, continuation, and failure records of $T_{\text{exp}}[\bar{O}]$;
3. Match_{ν}^{D5} survives activity reduction and lies on a target-relative carrier path ending in the actual neutrino carrier, mass-response, or matching architecture represented by H ; it is not merely a shared charge label, common-apex tag, or disconnected auxiliary sector;
4. the operator or process, the D5 neutrino-matching incidence, and the retained base source admit one simultaneous physical apex satisfying the pure, cross, matching,

continuation, and common-cone equations of D5-03;

5. for every retained oscillation evaluator E and licensed old continuation u ,

$$E(u(W_{H,\mathcal{O}})) = E(u(W_H)).$$

Equivalently, the target-bearing D5 incidence is physically active at the D5 target while projecting to zero in the complete retained oscillation standing algebra.

Distinct nonzero source magnitudes are not standing-distinct unless an admitted D5 evaluator separates them.

Define

$$\mathcal{O}_{D5}^{\text{tr}} := \{\mathcal{O} : \forall q \in Q_{\text{osc}}^{\text{SCIC}} \exists H \in \mathcal{N}_{\text{osc}}[q] \text{ with } \text{NMTr}_{D5}(\mathcal{O}; H) = 1\}.$$

This class is a property of already admitted physical lifts, not a rule for enlarging the source category. The stated D5 comparison class does not establish that $\mathcal{O}_{D5}^{\text{tr}}$ is nonempty, so its universal-membership status is NotReady. A physically valid $B - L$ -violating sector with no D5 neutrino-matching incidence is a target-inert PCR-3 spectator and cannot witness membership in $\mathcal{O}_{D5}^{\text{tr}}$.

4.5 Normal forms

For each target tag τ , a source normal form contains:

1. a B15 source core H ;
2. a target-tagged action and charge packet;
3. all applicable operator, anomaly, global, and breaking records;
4. source equations and activity predicates;
5. matching, continuation, protector/violation, and bearer incidences;
6. typed null terminals for inactive target sorts;
7. provenance and a complete status frontier.

Lawful arrows are base source isomorphisms, fixed-boundary gauge and basis transformations, charge-lattice automorphisms, target-preserving normalization maps, family permutations preserving the source trace, vacuum-orbit and stabilizer isomorphisms, global-packet isomorphisms, and source-chart maps. A charge rescaling is lawful only when the charge lattice and every coupling and response record transform contragrediently.

4.6 Framework source groupoids

For a B15 frame H_a and target τ , let

$$\mathcal{G}_{\tau,a}^{\text{src}} = [\Theta_{\tau,a}^{\text{coh}} / \Gamma_{\tau,a}^{\text{src}}],$$

where $\Theta_{\tau,a}^{\text{coh}}$ is the exact source-coherent, activity-reduced solution object and $\Gamma_{\tau,a}^{\text{src}}$ is its lawful presentation groupoid. The total source atlas is the Grothendieck construction

$$\mathcal{A}_{D5}^{\text{src}}(\tau) = \int_{a \in \text{B15}} \mathcal{G}_{\tau,a}^{\text{src}},$$

with pairwise and higher overlaps taken over the concrete base source core and the target tag.

Theorem 4.4 (D5-02 – Source authorization and complete candidate groupoid). *For every $\tau \in \mathcal{F}_{D5}$ and every B15 frame H_a , $\mathcal{G}_{\tau,a}^{\text{src}}$ is source-complete for the declared finite semantic comparison class. Every semantic D5 source realization has a source-coherent normal form; every generated coherent normal form has a semantic interpretation; and two points are isomorphic exactly when their complete activity-reduced source traces agree up to lawful D5 redescription. The reduct satisfies*

$$U_{D5} \circ I_{D5} = \text{id}_{\text{Src}_{D5}^{\text{base}}}.$$

Proof. The result is the framewise application of the SLR–D5–1 source-representation theorem. A finite semantic realization determines a finite typed hypergraph of actions, charge objects, anomaly records, operator incidences, global data, breaking data, continuation relations, and bearer incidences. Each semantic primitive is represented by a source sort or relation, and source coherence is precisely the conjunction of the action, charge, anomaly, operator, stabilizer, remnant, matching, and incidence laws. This yields a normal form. Conversely, the source equations provide the physical interpretation of every coherent normal form. Activity reduction removes only source-null D5 incidences, and trace extensionality gives the equivalence statement. By definition, I_{D5} adds only target-typed null D5 data and U_{D5} removes them, so their composite is the identity on $\text{Src}_{D5}^{\text{base}}$. $\square$

4.7 Complete complement decomposition

The raw parameter object decomposes into disjoint target-tagged strata:

$$\Theta_{\tau,a}^{\text{raw}} = Z_{\tau,a}^{\text{src}} \sqcup F_{\tau,a}^{\text{fail}} \sqcup U_{\tau,a}^{\text{undef}} \sqcup N_{\tau,a}^{\text{N/A}} \sqcup R_{\tau,a}^{\text{ROOS}} \sqcup S_{\tau,a}^{\text{upd}} \sqcup T_{\tau,a}^{\text{upd}}.$$

An active failure is never reclassified as inapplicable; an unavailable global packet is not interpreted as trivial global form; and a malformed remnant is not treated as a broken symmetry with an unknown quotient. This total decomposition underlies the fibre-exhaustion theorem.

5 Simultaneous physical realization

5.1 From source coherence to physical survival

A source normal form may be well typed and nevertheless fail anomaly consistency, vacuum compatibility, global completion, matching, or common-cone realization. D5 therefore passes from the source category to a semantic physical category before quotienting by carrier identity, in accordance with the one-point survival discipline of PCR-4 [2].

Let W_H be a base physical realization and let $S \in \mathcal{G}_{\tau,a}^{\text{src}}$ be a D5 source point. If $B_{D5}(W_H)$ denotes the inherited boundary and source-attachment interface, the enriched object is the split-anchored amalgam

$$W_{H,S} = W_H \cup_{B_{D5}(W_H)}^{\text{phys}} D_{D5}(H, S),$$

where $D_{D5}(H, S)$ carries the D5 action, charge, anomaly, global, breaking, matching, response, and bearer data.

5.2 Residual decomposition

The physical residual system has four parts:

- $\mathcal{E}_{D5}^{\text{inh}}$: inherited PCR-4 and base equations,
- $\mathcal{E}_{D5}^{\text{pure}}$: D5 action, charge, anomaly, breaking, and global equations,
- $\mathcal{E}_{D5}^{\text{cross}}$: shared-field, matching, covariance, and bearer incidences,
- $\mathcal{E}_{D5}^{\text{cone}}$: one-apex identity and compatibility.

The physical model locus is

$$\text{PhysMod}_{D5}(\tau) = \{W_{H,S} : \mathcal{E}^{\text{inh}} = \mathcal{E}^{\text{pure}} = \mathcal{E}^{\text{cross}} = \mathcal{E}^{\text{cone}} = 0\}.$$

5.3 Interpretation of the D5 source data

The added source sorts have the following physical meanings.

- SymAction is represented by a faithful local, global, accidental, broken, or remnant action on the specified bearer.
- ChargeObj is represented by the integral or rational charge lattice and its faithful quotient.
- AnomalyRecord is represented by the complete applicable coefficient or global-obstruction packet at the declared probe scope.
- Break is represented by a source-active operator or process, or by a physical breaking carrier and vacuum orbit.
- Stabilizer and Remnant are reconstructed from the same action and vacuum branch.
- Protect ranges over the complete target-active violation and continuation grammar; it is not a parameter value.
- BearerIncidence is represented by common continuation-bearing carrier references, spectral subobjects, or higher-arity attachment data.

Theorem 5.1 (D5-03 – Constraint-preserving physical symmetry realization). *Every source point in $Z_{\tau,a}^{\text{src}}$ has exactly one of two dispositions:*

1. *there exists a semantic physical apex $W_{H,S} \in \text{PhysMod}_{D5}(\tau)$ with*

$$\chi_{\text{car}}(W_{H,S}) \simeq P_{H,S}^{\text{red}}, \quad \text{Res}_{\text{base}}(W_{H,S}) \simeq W_H;$$

2. *a named residual in $\mathcal{E}^{\text{pure}}$, $\mathcal{E}^{\text{cross}}$, or $\mathcal{E}^{\text{cone}}$ fails, producing a typed physical complement.*

No accepted D5 point is assembled from records belonging to different physical apices.

Proof. The B15 base realization already satisfies the inherited residuals. The D5 source point contributes a finite diagram of additional physical uses. Forming the pushout along the fixed base boundary and imposing the D5 constitutive equations produces a single simultaneous system. If its pure, cross, and cone equations have a common solution, the result is a semantic physical model whose activity-reduced carrier trace agrees with the source trace. If no common solution exists, the finite residual decomposition identifies a first failing equation or an undefined common-cone condition. Because all records share the same apex and overlap maps, records from different physical models cannot be combined to satisfy the cone equations. $\square$

5.4 Target-specific physical conditions

The residual equations specialize as follows.

- At $T_{\text{acc}}^{(4)}$, every source-active $d \leq 4$ monomial has zero $B - L$ charge.
- At $T_{\text{bg}, \text{min}}$, the chiral spectrum is the minimal one and the complete anomaly tuple is evaluated in the fixed convention.
- At $T_{\text{loc}, 3N}$, the local action is faithful, the chiral anomaly-cancelling packet contains one active ν_R per family, every active interaction is covariant, and the complete local anomaly tuple vanishes. Additional anomaly-neutral vectorlike support is admissible only when it preserves these records.
- At T_{glob} , every active source operation is invariant and the protector remains valid under all licensed continuations.
- At $T_{\text{exp}}[\mathcal{O}]$, the operator or process is non-null and physically instantiated; its source-active D5 neutrino-matching/bearer incidence reaches the actual neutrino carrier, mass-response, or matching architecture of the base point; and that incidence, its continuation, and its failure semantics coexist on the same physical apex. A physically valid $B - L$ -violating sector with no such target-bearing path is a spectator and fails the D5 cross/-matching residual rather than defining an explicit-breaking lift.
- At $T_{\text{sp}}[q_\Phi]$, the pre-breaking source is invariant, the vacuum lies in a declared nonzero orbit, and the stabilizer is computed from the actual action.
- At $T_{\text{rem}}[\Gamma, q_\Phi]$, the residual action is the faithful quotient of the stabilizer and charge lattice, with every activated global record present.

5.5 Failure localization

A physical failure is recorded by

$$\text{FailCert}_\tau(H, S) = (\tau, a, \theta, \mathcal{E}_j, \text{scope}, \text{Prov}, \text{first separator}).$$

This object is invariant under lawful source arrows. An anomaly failure cannot be removed by charge normalization, a remnant failure cannot be relabelled as spontaneous breaking, and a missing global packet cannot be replaced by a local polynomial record.

6 Restriction to the oscillation quotient

6.1 Evaluation with D5 structure present

A D5 realization restricts to a base carrier only if the oscillation target remains evaluable while the new symmetry structure is physically present, as required by PCR-5C and the preceding D-series analyses [5, 6, 7]. Deleting D5 cells and recovering the base source is not sufficient: a new charge action, breaking sector, or global relation may modify an inherited continuation or matching channel before deletion.

Let Ev_{osc} denote the saturated family of retained oscillation evaluators and licensed continuations. The D5 physical algebra decomposes into the inherited subalgebra and the target-relative ideal generated by D5-only cells. Cross terms remain explicit rather than being assumed to vanish.

Definition 6.1 (D5 transparent locus). The transparent locus $\mathcal{V}_{\text{D5}}^{\text{tr}}(\tau)$ consists of physical

points $W_{H,S}$ such that:

1. every inherited PCR-4 equation holds on the enriched apex;
2. each D5-only generator is annihilated by every inherited evaluator after every licensed inherited continuation;
3. every cross residual contributing to an inherited packet channel vanishes;
4. the fixed current-source deletion field, activity record, matching orientation, boundary incidence, and same-source continuation are unchanged.

6.2 Channelwise equality

For $W_{H,S} \in \mathcal{U}_{D5}^{\text{tr}}(\tau)$, define

$$\Xi_{\text{osc},D5}^{\text{ret}}(W_{H,S}) = (T^{\text{ret}}, R^{\text{ret}}, \Delta^{\text{ret}}, C^{\text{ret}}, M^{\text{ret}}, B^{\text{ret}}, F^{\text{ret}}).$$

Transparency requires

$$\begin{aligned} T^{\text{ret}} &= T^{\text{face}}(W_H), & R^{\text{ret}} &= R^{\text{stand}}(W_H), \\ \Delta^{\text{ret}} &= \Delta_{\text{fr}}^{\text{del}}(W_H), & C^{\text{ret}} &= C^{\text{act}}(W_H), \\ M^{\text{ret}} &= M^{\text{match}}(W_H), & B^{\text{ret}} &= B^{\text{bdy}}(W_H), \\ F^{\text{ret}} &= F^{\text{same}}(W_H). \end{aligned}$$

Theorem 6.2 (D5-04 – Retained-oscillation transparency). *For every physical point entering a D5 lift fibre,*

$$\Phi_{4,D5}^{\text{ret}}(W_{H,S}) = \Phi_4(W_H), \quad \Xi_{\text{osc},D5}^{\text{ret}}(W_{H,S}) = \Xi_{\text{osc}}(W_H).$$

The equalities hold with D5 structure present. They are natural under lawful source and physical arrows.

Proof. The source-representation theorem supplies a split conservative reduct and a typed D5-only ideal. D5-03 evaluates the inherited, pure-D5, cross, and cone residuals on one physical object. On the transparent locus, every inherited evaluator annihilates the D5-only ideal, and every cross contribution to an inherited channel vanishes. The inherited evaluator values are therefore unchanged. Naturality follows because lawful arrows preserve the split, the inherited evaluator algebra, and the cross-residual equations. $\square$

6.3 Restriction functors

Define

$$R_\tau : \mathcal{N}_{D5}(\tau) \longrightarrow \mathcal{N}_{\text{osc}}$$

by sending a transparent enriched point to its retained oscillation standing core. Nontransparent points lie outside its domain and receive a typed failure or target-update status.

Theorem 6.3 (SCIC descent and no merger). *R_τ descends to*

$$\rho_\tau : \Pi_0 \mathcal{N}_{D5}(\tau) \longrightarrow Q_{\text{osc}}^{\text{SCIC}}.$$

If $q_1 \neq q_2$ in $Q_{\text{osc}}^{\text{SCIC}}$, no enriched classes restricting to q_1 and q_2 are identified in the D5 quotient.

Proof. Representative independence follows from channelwise equality and naturality un-

der lawful arrows. If two enriched points were identified while their base classes differed, enriched equivalence would force agreement of their complete retained packets, contradicting the fixed SCIC separator between q_1 and q_2 . $\square$

6.4 Intrinsic and transparent extensions

Some D5 structures are intrinsic: the action or breaking source already belongs to the base source architecture. Others are neutrino-matched transparent extensions: they add target-active D5 structure through a current neutrino-matching incidence while projecting to zero under every retained oscillation evaluator and continuation. These branches remain distinct in the fibre atlas. Transparency does not make the D5 structure physically inert; it means that the structure is active at the D5 target, genuinely attached to the neutrino carrier architecture, and nevertheless leaves the retained oscillation packet unchanged. A decoupled D5 sector is not a transparent lift but a spectator removed before D5 survival is evaluated.

7 Lift fibres and global reconstruction

7.1 Framewise solution objects

For a target τ and B15 frame a , let $\Theta_{\tau,a}^{\text{phys}}$ be the physical parameter object after source coherence, activity reduction, and simultaneous realization. Define

$$Z_{\tau,a} = \{\theta \in \Theta_{\tau,a}^{\text{phys}} : W_{H_a, S(\theta)} \in \mathcal{V}_{\text{D5}}^{\text{tr}}(\tau) \text{ and } \text{Acc}_{\tau} = 1\}.$$

The lawful groupoid $\Gamma_{\tau,a}$ acts on $Z_{\tau,a}$, producing the fibre chart $[Z_{\tau,a}/\Gamma_{\tau,a}]$.

7.2 Connected homotopy fibres

For $q \in Q_{\text{osc}}^{\text{SCIC}}$, define the PCR-5C connected homotopy fibre [5]

$$\mathcal{C}_{\tau}(q) = \mathcal{N}_{\text{D5}}(\tau) \times_{\mathcal{N}_{\text{osc}}}^h \mathcal{N}_{\text{osc}}[q].$$

This groupoid lies over the complete connected base component and retains every source presentation and lawful automorphism. Pairwise and higher overlaps are homotopy fibre products over the concrete retained standing core.

Theorem 7.1 (D5-05 – Exact lift-fibre exhaustion). *For every target τ and base class q , $\mathcal{C}_{\tau}(q)$ is canonically equivalent to the codescent of the exact framewise solution groupoids over the B15 charts representing q :*

$$\mathcal{C}_{\tau}(q) \simeq \text{hocolim}_{a \in \text{B15}(q)} [Z_{\tau,a}/\Gamma_{\tau,a}].$$

Every source-coherent physical D5 lift occurs in this object, every duplicate presentation is identified by a lawful overlap, and every excluded point belongs to a typed complement.

Proof. D5-02 gives the complete source atlas. D5-03 sends each source point either to a simultaneous physical realization or to a typed physical failure. D5-04 restricts precisely the transparent accepted points to the connected base core. PCR-5B overlap closure and PCR-5C homotopy-correct fibre reconstruction identify duplicate charts while retaining isotropy and standing separators. The total status decomposition covers the complement. $\square$

7.3 Fibrewise assembly of the enriched quotient

The enriched quotient is recovered fibrewise:

$$Q_{D5}(\tau) \cong \coprod_{q \in Q_{osc}^{SCIC}} \pi_0 \mathcal{L}_\tau(q),$$

and

$$\text{im } \rho_\tau = \{q \in Q_{osc}^{SCIC} : \mathcal{L}_\tau(q) \neq \emptyset\}.$$

This is an exact symbolic reconstruction, not a finite sample. Sections 9–11 identify the images with invariant source-semantic subatlases.

7.4 Fibre geometry

Each connected fibre receives one of the coarse types

Empty, Stable, Finite, Countable, Continuous, Functional, Categorical, Mixed, NotReady.

The type is assigned after lawful quotienting. A continuous raw parameter may be normalization skin and disappear, whereas a finite source presentation may still yield functional or categorical moduli through internal structure and automorphisms.

7.5 Failure localization

If $\mathcal{L}_\tau(q) = \emptyset$, the fibre carries the separator object

$$\text{Sep}_\tau(q) = (\tau, q, \text{all frames, first active failures, overlap closure, Prov}).$$

A fibre is empty only when every B15 representative and every overlap has failed or changed scope. Failure of a single benchmark does not eliminate a base class.

8 Charge lattices and anomaly descent

8.1 Charge objects and lawful equivalence

A D5 charge object is the tuple

$$\bar{Q} = (\Lambda, Q, \ker Q, \text{unit, family, quot, act}),$$

where Λ is the integral or rational charge lattice, Q the action homomorphism, $\ker Q$ the ineffective subgroup, unit the conventional charge unit, family the family index, quot the faithful quotient data, and act the target-active support. Displayed charge numbers do not by themselves determine this object.

A lawful charge transformation is an isomorphism $u : \Lambda \rightarrow \Lambda'$ together with the induced contragredient transformation of every action, coupling, operator, anomaly, and background record. A numerical rescaling without the corresponding transformation of the action and lattice is not an equivalence.

8.2 Target-indexed anomaly functors

For a source point H , define

$$\mathfrak{A}_\tau(H) = (A_{SU(3)^2-X}, A_{SU(2)^2-X}, A_{Y^2-X}, A_{Y-X^2}, A_{\text{grav}^2-X}, A_{X^3})_\tau,$$

where $X = B - L$ and the target tag τ is retained. Each coefficient is the complete left-handed Weyl sum with representation multiplicity and the fixed chirality convention. Global anomalies, discrete anomalies, and extended-sector responses have distinct codomains and are not coerced into this tuple.

Lemma 8.1 (Normalization covariance). *Under a lawful rescaling $X \mapsto rX$ and the corresponding transformation of the background/gauge coupling, the vanishing or nonvanishing status of each applicable anomaly row is invariant. The coefficient tuple transforms homogeneously with the degree of the charge occurrence.*

Proof. Mixed coefficients linear in X scale by r , the $Y - X^2$ coefficient by r^2 , and the cubic coefficient by r^3 . The transformed coupling and charge unit preserve the physical obstruction. Since $r \neq 0$, vanishing is unchanged. $\square$

Lemma 8.2 (Family permutation invariance). *Permuting identical family copies is a lawful source automorphism and leaves every complete anomaly coefficient unchanged.*

Proof. Each coefficient is a sum over the complete family-indexed chiral spectrum, and a permutation only reorders the summands. $\square$

8.3 Minimal and three-singlet coefficients

In the left-handed convention of SYM-7 [12], the minimal chiral spectrum has vanishing mixed anomalies with the dynamical Standard Model gauge factors, while

$$A_{\text{grav}^2-(B-L)} = -N_f, \quad A_{(B-L)^3} = -N_f.$$

For $N_f = 3$, both coefficients are -3 . Adding one right-handed neutrino per family, represented by a left-handed conjugate of charge $+1$, changes both sums by $+N_f$ and makes them vanish.

Theorem 8.3 (D5-06 – Charge-lattice and anomaly descent). *The complete D5 charge/action/anomaly packet descends through the lawful presentation groupoid. Every surviving distinction is exactly one of:*

1. *invariant charge-lattice or faithful-quotient content;*
2. *an invariant local/background/global obstruction record at its target scope;*
3. *a target update caused by a changed charge lattice, probe, global form, or operator category.*

No family relabeling, basis change, or lawful normalization creates an additional D5 class.

Proof. The charge lattice and action are functorial under lawful arrows. Operator charge balance is preserved because every field and coefficient transforms within the same source diagram. The anomaly functors are sums of representation-theoretic invariants; they are covariant under normalization and invariant under basis and family permutations. A transformation that changes the faithful quotient, probe set, or global packet changes the target

rather than its presentation. The quotient therefore removes presentation skin while preserving every target-bearing obstruction. $\square$

8.4 No partial anomaly closure

Corollary 8.4. *At $T_{\text{bg},\text{min}}$, vanishing of the four dynamical mixed-gauge rows does not satisfy the target because the cubic and mixed-gravitational rows are active and nonzero. At $T_{\text{loc},3N}$, local cancellation is established only after every applicable row and every source-covariance condition passes.*

8.5 Global-form dependence

When extended operators, monopoles, line spectra, fractionalization, higher groups, or defects are target-active, the source point carries the SYM-9 global-completion packet

$$\mathfrak{G}_\Gamma = (G_\Gamma, \mathcal{P}_\Gamma, \mathcal{L}_\Gamma, \mathcal{S}_\Gamma, \mathcal{M}_\Gamma, \mathcal{A}_\Gamma, \Theta_\Gamma, \Phi_\Gamma^{\text{phys}}, \mathcal{F}_\Gamma, \mathcal{H}_\Gamma, \mathcal{N}_\Gamma).$$

Statements at this scope are fixed-target or explicitly family-indexed. A sharp-local result intentionally omits these discriminants and cannot later be cited as a completed global theorem.

9 $B - L$ targets: accidental, background, local, and global

9.1 Invariant predicates on the base atlas

For a B15 representative H , define the following target-separated predicates using the $B - L$ results of SYM-7 and SYM-9 [12, 14].

Definition 9.1 (Renormalizable accidental predicate). $\text{Acc}_{\leq 4}(H)$ holds when every source-active operator O in the declared $d \leq 4$ D5 envelope satisfies $\Delta(B - L)(O) = 0$. Higher-dimensional operators remain part of the source provenance but do not enter this bounded predicate.

Definition 9.2 (Minimal-background predicate). $\text{MinBG}(H)$ holds when the target-active $B - L$ chiral spectrum is exactly the minimal Standard Model spectrum at the fixed family number and chirality convention, the background probe is present, and the complete anomaly tuple equals the minimal packet.

Definition 9.3 (Local three-singlet predicate). $\text{Loc}_{3N}(H)$ holds when H admits a faithful unbroken local $U(1)_{B-L}$ action with one source-active right-handed neutrino per family in the chiral cancellation packet, every active source operation is covariant, and the complete applicable local polynomial anomaly tuple vanishes. Additional anomaly-neutral vectorlike support is allowed when it preserves the target action, anomaly packet, and neutrino matching.

Definition 9.4 (Exact global protector predicate). $\text{GlobProt}(H)$ holds when H admits a faithful global $B - L$ action on every target-active cell together with a protector relation that remains valid under every licensed D5 continuation, thereby excluding every active $\Delta(B - L) \neq 0$ route in the fixed target grammar.

These predicates are invariant under the lawful B15/D5 groupoids and hence define subatlases of $Q_{\text{osc}}^{\text{SCIC}}$:

$$Q_P := \{q \in Q_{\text{osc}}^{\text{SCIC}} : \exists H \in \mathcal{N}_{\text{osc}}[q] \text{ with } P(H)\}.$$

9.2 Accidental conservation through dimension four

Theorem 9.5 (Accidental-fibre image).

$$\text{im } \rho_{\text{acc}}^{(4)} = Q_{\text{Acc} \leq 4}.$$

This image is nonempty and proper. It contains the minimal/Weinberg law-only marked sector and the exact-Dirac sector, while it excludes every representative whose active renormalizable source includes a Majorana mass or another $d \leq 4$ $\Delta(B-L) \neq 0$ term.

Proof. A lift at $T_{\text{acc}}^{(4)}$ is accepted exactly when the complete active $d \leq 4$ operator packet is invariant, which is the predicate $\text{Acc} \leq 4$. Since the condition is presentation invariant, existence over a connected base class is precisely membership in $Q_{\text{Acc} \leq 4}$. A Weinberg law-only realization contains no $d \leq 4$ $B-L$ violation and supplies a witness. A full-rank type-I realization with an active renormalizable Majorana term contains a dimension-three M_R source with $\Delta L = 2$ and fails. Both are lawful PCR-4/PCR-5B base classes. $\square$

Corollary 9.6 (Target-bound nature of accidental standing). *The Weinberg EFT can belong to $Q_{\text{Acc} \leq 4}$ while failing Q_{GlobProt} . Accidental conservation through $d \leq 4$ therefore does not determine the exact-global image.*

9.3 The minimal background anomaly packet

Theorem 9.7 (Minimal-background image).

$$\text{im } \rho_{\text{bg,min}} = Q_{\text{MinBG}}.$$

This image is nonempty and proper. Its accepted points carry the nonzero cubic and mixed-gravitational background coefficients; those coefficients are positive standing records of obstruction, not target failures.

Proof. The target fixes the minimal chiral inventory and the background probe. A point lifts exactly when a base representative has that inventory and carries the complete minimal anomaly packet. The minimal law-only/Weinberg source provides such a representative. A three- ν_R source changes the target-active chiral inventory and cancels the two background deficits; it therefore belongs to a different target rather than to the minimal-background fibre. $\square$

9.4 Unbroken local $U(1)_{B-L}$ with three singlets

Theorem 9.8 (Local $3\nu_R$ image).

$$\text{im } \rho_{\text{loc},3N} = Q_{\text{Loc}_{3N}}.$$

The image is nonempty and proper. It includes full-rank Dirac three-singlet realizations and excludes active Weinberg and renormalizable-Majorana source support at the unbroken local target.

Proof. The target requires a faithful local action, one chiral ν_R per family in the anomaly-cancelling packet, covariant target-active support, and a vanishing local anomaly polynomial. These are precisely the clauses of Loc_{3N} . A three-singlet Dirac realization carrying this faithful local action and no source-active $\Delta(B-L) \neq 0$ incidence satisfies the target; local covariance in particular excludes an M_R source at the unbroken target. An active Weinberg oper-

ator has $\Delta L = 2$ and is not covariant under unbroken $U(1)_{B-L}$, and a nonzero renormalizable Majorana source has the same obstruction. Their base classes therefore have empty local fibres unless a breaking target is declared. Additional anomaly-neutral vectorlike sectors do not change the chiral cancellation packet and remain admissible when they preserve the complete target records. $\square$

Corollary 9.9 (Local-coupling standing-authority boundary). *At the frozen $T_{\text{loc},3N}$ target, the existence of a continuous positive source parameter g_{BL} does not by itself imply continuum many standing-distinct lifts over a fixed base class. Source activity distinguishes the null and non-null local action, while further distinction among positive coupling magnitudes requires a separately frozen target-admitted response evaluator. No such universal quantitative g_{BL} evaluator is part of the present target packet. Accordingly,*

$$|\pi_0 \mathcal{L}_{\text{loc},3N}(q)| \geq \mathfrak{c}$$

is not inferred from variation of g_{BL} alone.

Proof. The source rule retains every non-null target-active local action without a magnitude threshold, while standing is determined by the admitted action, charge, anomaly, matching, continuation, and failure records. A positive coupling magnitude is therefore a source parameter unless an admitted D5 evaluator makes its numerical variation standing-visible. The proper-continuum conclusion for $Q_{\text{Loc},3N}$ is unaffected because Theorem 9.11 supplies continuum many source-coherent physical base classes whose heavy threshold and oriented matching data already carry standing. $\square$

9.5 Exact global protection

Theorem 9.10 (Exact-global image).

$$\text{im } \rho_{\text{glob}} = Q_{\text{GlobProt}}.$$

The image is nonempty and proper. It contains exact-Dirac and other source architectures with a complete continuation-stable $B - L$ protector and excludes active Weinberg and Majorana source support.

Proof. A global lift must furnish a faithful action and a protector quantified over the complete target-active source and continuation grammar; this is exactly GlobProt. A Dirac three-singlet realization carrying the faithful exact $B - L$ action and the continuation-stable Protect relation provides a nonempty witness. Source covariance excludes M_R and every other $\Delta(B - L) \neq 0$ incidence throughout the licensed continuation grammar. Thus $M_R = 0$ is a consequence of the protected source architecture rather than the evidence for protection. By contrast, a Weinberg or Majorana source contains an active violation and cannot satisfy the protector. A later coefficient choice cannot repair that source failure. $\square$

9.6 Exact physical-response vectorlike Dirac threshold continuum

Theorem 9.11 (Source-realized exact-response vectorlike Dirac threshold continuum). *Fix a full-rank target-admissible light Dirac spectral package*

$$M_D^* = U \Sigma V^\dagger, \quad \Sigma = \text{diag}(\sigma_1, \sigma_2, \sigma_3), \quad \sigma_i > 0,$$

and let $E_{\text{osc}}^{\text{max}}$ be the largest kinematic energy in the frozen oscillation-boundary record. Choose

$$\Lambda > E_{\text{osc}}^{\text{max}}, \quad 0 < s < \min\left\{\frac{1}{4}, \frac{\nu}{8\sqrt{2}\Lambda}\right\}, \quad c = \sqrt{1 - s^2},$$

so that the fixed event-complete light response with charged-current matrix cU , light singular values Σ , and its associated null/no-event record lies in the selected B15 survivor chart. For $\lambda \in I_* = (1, 2)$ set $H_\lambda = \lambda\Lambda$ and define

$$\begin{aligned} Y_\nu &= \frac{\sqrt{2}}{\nu} cU\Sigma V^\dagger, & Y_{LN,\lambda} &= -\frac{\sqrt{2}}{\nu} sH_\lambda U, \\ \mu &= s\Sigma V^\dagger, & M_{N,\lambda} &= cH_\lambda \mathbf{1}. \end{aligned}$$

Let N_L, N_R be equal-rank Standard-Model-singlet multiplets of hypercharge zero and conventional $B - L = -1$. Then the source terms

$$-\bar{L}\widetilde{H}Y_\nu\nu_R - \bar{L}\widetilde{H}Y_{LN,\lambda}N_R - \overline{N_L}\mu\nu_R - \overline{N_L}M_{N,\lambda}N_R + \text{h.c.}$$

together with the protected exact-Dirac three- ν_R source data define one source-coherent PCR-4/B15 physical apex $\times_\lambda$ for every $\lambda \in I_*$. After electroweak symmetry breaking the complete Dirac mass map is

$$\mathbb{M}_\lambda = \begin{pmatrix} cU\Sigma V^\dagger & -sH_\lambda U \\ s\Sigma V^\dagger & cH_\lambda \mathbf{1} \end{pmatrix} = \mathcal{U}_L \begin{pmatrix} \Sigma & 0 \\ 0 & H_\lambda \mathbf{1} \end{pmatrix} \mathcal{U}_R^\dagger,$$

where

$$\mathcal{U}_L = \begin{pmatrix} U & 0 \\ 0 & \mathbf{1} \end{pmatrix} \begin{pmatrix} c\mathbf{1} & -s\mathbf{1} \\ s\mathbf{1} & c\mathbf{1} \end{pmatrix}, \quad \mathcal{U}_R = \begin{pmatrix} V & 0 \\ 0 & \mathbf{1} \end{pmatrix}.$$

Consequently

$$\mathcal{U}_L^\dagger \mathbb{M}_\lambda \mathcal{U}_R = \text{diag}(\Sigma, H_\lambda \mathbf{1})$$

for every λ . At the source matching chart, the exact light pole spectrum, canonical kinetic normalization, light charged-current boundary matrix cU , heavy charged-current boundary matrix $-sU$, and every target-energy light propagation amplitude are independent of λ ; direct heavy production remains in the same kinematically unavailable status because $H_\lambda > E_{\text{osc}}^{\text{max}}$. The physical heavy threshold is

$$\text{sv}_{\text{heavy}}(\mathbb{M}_\lambda) = H_\lambda \mathbf{1}.$$

The family is exactly local- $B - L$ covariant and carries the source-level exact-global protector. The complete licensed B15 matching/RG/EFT continuation and the resulting SCIC separation are proved in Theorem 9.12; neither is inferred from the finite-dimensional factorization alone.

Proof. All four displayed source operators are Lorentz and $SU(2)_L \times U(1)_Y$ invariant. With

$$L \sim (\mathbf{2}, -\tfrac{1}{2})_{-1}, \quad \widetilde{H} \sim (\mathbf{2}, -\tfrac{1}{2})_0, \quad \nu_R, N_L, N_R \sim (\mathbf{1}, 0)_{-1},$$

they are also exactly $B - L$ covariant. In a left-handed anomaly convention the added vectorlike sector contributes N_L of charge -1 and N_R^c of charge $+1$; its linear and cubic contributions cancel, and all mixed Standard-Model contributions vanish because both fields are gauge singlets. The original three chiral ν_R remain the family-indexed local anomaly-cancelling packet.

The two matrices $\mathcal{U}_L$ and $\mathcal{U}_R$ are unitary and independent of λ . Direct multiplication gives the displayed factorization, so the singular values of the complete physical mass operator are

exactly

$$\{\sigma_1, \sigma_2, \sigma_3, H_\lambda, H_\lambda, H_\lambda\}.$$

The diagonalization is unitary on both chiral kinetic spaces; hence no wave-function renormalization or noncanonical residue is being suppressed. The active rows of $\mathcal{U}_L$ give the fixed light and heavy charged-current matrices cU and $-sU$. At all energies belonging to the frozen oscillation record the heavy modes are kinematically unavailable. The light boundary-to-boundary amplitude is therefore

$$\mathcal{A}_{\beta\alpha}^{(\lambda)}(L, E) = c^2 \sum_i U_{\beta i} \exp\left(-\frac{i\sigma_i^2 L}{2E}\right) U_{\alpha i}^*,$$

together with the same source-chart event-complete null/no-event record, for every λ . The frozen response modes are represented by nondegenerate finite intervals and the source/detector normalization domain is fixed before carrier evaluation; choosing an interior exact-Dirac benchmark and then taking $s > 0$ sufficiently small supplies the stated source-scale physical packet. This is exact physical-response control at the source matching chart, not an identification of a Schur complement with the light pole mass. The claim that the full response, normalization, mode-class, applicability, and failure records remain in one accepted B15 survivor chart is supplied only by Theorem 9.12.

The source graph is target-active: $s > 0$ makes both the $L-N_R$ and $N_L-\nu_R$ incidences non-null, while the heavy states have the fixed nonzero charged-current projection $-sU$. Hence the vectorlike sector is not removed by spectator stripping. The bound on s gives $\|Y_{LN,\lambda}\|_2 < 1/4$ throughout I_* , and all remaining source matrices are finite. The scalar potential is unchanged, the electroweak vacuum remains a solution at vanishing fermion background, and the exact unitary factorization supplies one positive physical propagation/matching object. Thus the gauge, anomaly, vacuum, boundary, and source-scale common-cone records coexist on one source-coherent physical apex. The licensed continuation and final failure/applicability records are completed in Theorem 9.12.

Every active source incidence is exactly $B - L$ covariant, one source-active ν_R per family remains present, and the vectorlike sector is anomaly neutral. Thus the source-scale local-action and anomaly clauses hold. The faithful exact global action extends to all four covariant source operators and introduces no $\Delta(B - L) \neq 0$ route. The continuation-stable local/global acceptance clauses and the final SCIC separation are completed by Theorem 9.12.

Lawful kinetic-term-preserving source and mass-basis transformations already preserve the singular-value threshold at the source matching chart. The next theorem proves that the varying threshold remains the unique standing-visible difference after the complete licensed B15 continuation. $\square$

Theorem 9.12 (Standing-preserving finite-order RG/EFT matching for the exact-response continuum). *Let $\chi_\lambda^{(0)}$ be the source-formed exact-response family of Theorem 9.11. Fix the finite B15 EFT envelope through order K_{EFT} , choose $\lambda_0 \in I_*$, and let $\mu_{\text{light}}^{\text{max}}$ be the largest scale appearing in any target-retained light continuation. Refine the otherwise free choice of Λ so that*

$$\Lambda > \max\{E_{\text{osc}}^{\text{max}}, \mu_{\text{light}}^{\text{max}}\}.$$

After quotienting the finite light operator space by integration by parts, equations of motion, Ward identities, and lawful field redescrptions, let

$$\mathcal{B}_{\text{light}} = \{\mathcal{O}_{a,n} : 1 \leq a \leq N_n, 0 \leq n \leq K_{\text{EFT}}\}$$

be an independent gauge- and exact- $B - L$ -covariant basis dual to the independent continuous primitive B15 coordinates $r_{a,n}$. At each order, finite matching has the triangular form

$$r_{a,n}^{\text{match}}(\lambda) = p_{a,n}^{\text{src}}(\lambda) + \Delta r_{a,n}(\lambda; p_{<n}^{\text{src}}(\lambda)),$$

where the current-order independent source coefficient enters its defining renormalization condition with unit coefficient and the threshold term depends only on source data already fixed at lower order. For the accepted benchmark $r_{a,n}^*$, define before target evaluation the explicit recursive right inverse

$$p_{a,n}^{\text{src}}(\lambda) := r_{a,n}^* - \Delta r_{a,n}(\lambda; p_{<n}^{\text{src}}(\lambda)), \quad 0 \leq n \leq K_{\text{EFT}}.$$

Equivalently, after assembling the finite coefficient vectors,

$$C^{\text{match}}(\lambda) = C^{\text{src}}(\lambda) + \Delta C(\lambda) = C^*.$$

The source-to-matched map is triangular with unit diagonal in the independent quotient basis; no Jacobian regularity, local-surjectivity assumption, or implicit-function theorem is used. A source-formed heavy on-shell source counterterm simultaneously fixes the physical heavy pole to

$$H_\lambda = \lambda \Lambda.$$

Let $\mathcal{R}_{\text{light}}^{\text{cont}}$ be the resulting independent continuous light packet, and let $\mathcal{S}_{\text{light}}^{\text{disc}}$ contain the activity pattern, applicability, Pass/Fail/N/A/Undef/ROOS class, response-mode combinatorics, and categorical continuation/failure incidence. There exists a nondegenerate interval $I_{\text{match}} \subseteq I_*$ containing λ_0 such that

$$H_{\min} := \inf_{\lambda \in I_{\text{match}}} H_\lambda > \mu_{\text{light}}^{\max}$$

and no source-activity, threshold, response-mode, applicability, or failure frontier is crossed. The completed points $\widehat{x}_\lambda$, fixed before target evaluation, satisfy

$$\mathcal{R}_{\text{light}}^{\text{cont}}(\widehat{x}_\lambda) = \mathcal{R}_{\text{light}}^{\text{cont},*}, \quad \mathcal{S}_{\text{light}}^{\text{disc}}(\widehat{x}_\lambda) = \mathcal{S}_{\text{light}}^{\text{disc},*},$$

and their complete licensed continuation decomposes as

$$\text{Cont}_{\text{B15}}(\widehat{x}_\lambda) = (\text{Cont}_{\text{light}}^*, \text{Match}_{H_\lambda}, \text{Cont}_{\text{heavy},\lambda}).$$

For every primitive retained oscillation witness w , every licensed target-facing continuation u below $H_{\min}$, and all $\lambda, \mu \in I_{\text{match}}$,

$$w(u(\widehat{x}_\lambda)) = w(u(\widehat{x}_\mu)).$$

Every target-retained packet channel outside M^{match} is constant; within M^{match} , the common low-energy image is fixed and the standing-visible invariant that varies is the physical threshold H_λ with its oriented matching arrow. Every $\widehat{x}_\lambda$ is therefore an accepted B15 survivor satisfying both Loc_{3N} and GlobProt , and

$$\lambda \neq \mu \implies [\widehat{x}_\lambda]_{\text{SCIC}} \neq [\widehat{x}_\mu]_{\text{SCIC}}.$$

Hence

$$|Q_{\text{Loc}_{3N}} \cap Q_{\text{GlobProt}}| \geq \mathfrak{c}.$$

Proof. The continuous target packet is formed only after the standard finite-order quotient

by integration by parts, equations of motion, Ward identities, and lawful field redefinitions. Its coordinates are therefore independent physical source/matching coefficients rather than a redundant list of displayed Green-function entries. PCR-3 source saturation at the declared finite envelope supplies one exact- $B - L$ -covariant local source coefficient for every basis element $\mathcal{O}_{a,n}$, and the B15 evaluator reconstructs every retained continuous quantity from the dual coordinates $r_{a,n}$. Derived quantities constrained by Ward identities or canonical normalization are not counted as additional independent coordinates.

Fix one gauge- and exact- $B - L$ -covariant renormalization prescription and one threshold-matching convention across the family. At perturbative order n , the coefficient $p_{a,n}^{\text{src}}$ enters the corresponding independent renormalization condition linearly with coefficient one, while loop insertions and lower-order counterterms contribute the already determined function $\Delta r_{a,n}(\lambda; p_{<n}^{\text{src}})$. This gives the displayed triangular matching equation. Starting at $n = 0$ and proceeding recursively to K_{EFT} , define

$$p_{a,n}^{\text{src}}(\lambda) = r_{a,n}^* - \Delta r_{a,n}(\lambda; p_{<n}^{\text{src}}(\lambda)).$$

Substitution proves $r_{a,n}^{\text{match}}(\lambda) = r_{a,n}^*$ at every order. In the ordered independent basis the source-to-matched matrix is block triangular with identity diagonal blocks. Thus the matching rank and the right inverse are proved algebraically; no Jacobian or implicit regularity claim remains. The assembled vector equation is $C^{\text{match}} = C^{\text{src}} + \Delta C = C^*$.

The same finite-order recursion is applied to the independent heavy on-shell coordinate. If $\Delta H_n(\lambda)$ is the order- n heavy-pole shift after the lower-order source data are fixed, the source chart contains the order- n on-shell counterterm $h_n^{\text{src}}(\lambda) := -\Delta H_n(\lambda)$. Hence the physical heavy singular-value spectrum remains $H_\lambda \mathbf{1}$ through the frozen order.

The benchmark coefficients $r_{a,n}^*$, the heavy source family, every finite threshold contribution, every recursive compensating source coefficient, and the heavy on-shell counterterms are fixed as part of the lift functor before any target pass/fail result is consulted. They are not selected after evaluation and do not constitute same-act repair.

Separate continuous data from the discrete/categorical status object. The recursive right inverse fixes $\mathcal{R}_{\text{light}}^{\text{cont}}$ exactly. No member of $\mathcal{S}_{\text{light}}^{\text{disc}}$ is treated as an analytic variable. Choose the benchmark in the interior of one finite source-activity, vacuum, applicability, response-mode, and failure stratum. Required active coefficients are nonzero at the benchmark and vary continuously; coefficients forbidden by exact $B - L$ remain identically zero. Because the relevant activity zeros, threshold surfaces, response-mode changes, and applicability/failure frontiers are closed loci in the finite source chart, one may shrink to a nondegenerate interval I_{match} on which none is crossed and $H_{\min} > \mu_{\text{light}}^{\max}$. The complete discrete status packet is therefore constant on that interval.

The hypotheses of the standing-preserving EFT envelope and threshold-gluing theorem of SYM-10 [15] are now satisfied: the finite UV and IR envelopes and their order are fixed; the independent continuous packet supplies the overlap data; the explicit recursive right inverse supplies the operator-basis and Wilson-coefficient matching map; scheme and scale covariance are fixed; exact $B - L$, anomaly, and global-protector data are transported; and the B15 truncation/error packet is unchanged. Threshold gluing gives

$$\text{Cont}_{\text{B15}}(\widehat{\mathcal{X}}_\lambda) = (\text{Cont}_{\text{light}}^*, \text{Match}_{H_\lambda}, \text{Cont}_{\text{heavy},\lambda})$$

through the declared order. Above-threshold heavy dynamics may vary, but their only retained projection on the frozen oscillation target is the physical threshold and oriented matching arrow.

Every compensating operator and counterterm is gauge invariant and exactly $B - L$ covariant. The construction preserves the anomaly-neutral vectorlike pair, the three chiral ν_R cancellation packet, the continuation-stable exact-global protector, the common vacuum, the charged-current boundary, and the one-apex physical realization. Thus every completed point satisfies Loc_{3N} and GlobProt .

Finally, lawful source, basis, gauge, charge-normalization, scheme, and chart arrows preserve the physical heavy-pole spectrum and matching orientation. Since $H_\lambda \neq H_\mu$ for $\lambda \neq \mu$, the complete standing packets differ in M^{match} while agreeing in every other retained channel. Packet completeness gives distinct SCIC classes, and I_{match} has cardinality c . $\square$

Nonclaim 9.13 (Finite-order transport scope and fail-closed boundaries). The theorem is exact only at the frozen finite B15 EFT envelope and declared order. It asserts neither equality of above-threshold heavy observables nor equality outside that target. If an independent retained continuous coordinate lacks a dual source operator, or if the unit-diagonal triangular matching form fails after the Ward/EOM/field-redescription quotient, the construction returns $\text{NotReady}_{\text{MatchingBasisIncomplete}}$. If the interval crosses a source-activity, applicability, threshold, response-mode, or failure frontier, it returns $\text{NotReady}_{\text{ContinuationStratumCrossing}}$ rather than transporting the family across that boundary.

9.7 Proper-continuum strength

Theorem 9.14 (Proper-continuum marginal structure). *Each of $Q_{\text{Acc}_{\leq 4}}$, Q_{MinBG} , $Q_{\text{Loc}_{3N}}$, and Q_{GlobProt} contains at least continuum many SCIC classes, and each complement in $Q_{\text{osc}}^{\text{SCIC}}$ contains at least continuum many SCIC classes.*

Proof. Appendix C gives the marked source data and packet-rigidity details. The LAW family $C_{5,\lambda}/\Lambda_\lambda = \kappa_*$ varies a retained matching threshold while preserving the accidental and minimal-background predicates. Theorems 9.11 and 9.12 supply the source-realized $B - L$ -preserving vectorlike Dirac family and its standing-preserving continuation: they vary an anomaly-neutral vectorlike threshold while preserving the exact light pole spectrum, canonical charged-current response, complete target-retained low-energy RG/EFT packet, local cancellation, one-apex physicality, and exact-global protection. The PCR-5B active-singlet Majorana scaling family supplies a continuum outside all four images, and the LAW family supplies a continuum outside the unbroken local and exact-global images. In each case the varied threshold or matching datum is a standing coordinate rather than a lawful groupoid orbit. $\square$

Theorem 9.15 (D5-07 – Local/global/gaugeability frontier). *The four source predicates $\text{Acc}_{\leq 4}$, MinBG , Loc_{3N} , and GlobProt define four target-qualified acceptance functors with no coercions between them. Their exact restriction images are the four subatlases proved above. Mixed-gauge cancellation, cubic/mixed-gravitational background obstruction, local gaugeability, exact global protection, and globally complete gauged structure remain distinct records; none is obtained by deleting fields from another target packet.*

Proof. D5-06 proves charge and anomaly descent while retaining the probe and target tags. The four image theorems evaluate their source and physical packets separately. A morphism between two fibres would have to preserve the active operator envelope, chiral inventory, anomaly probe, action type, protector or global packet, and failure semantics. Whenever one of these coordinates differs, D5-01 identifies a target update. Local polynomial cancellation therefore cannot be promoted to exact-global or globally complete gauged standing, and background obstruction cannot be retyped as local failure. $\square$

The target-qualified conclusion is

$$\begin{aligned} \emptyset &\subsetneq Q_{\text{Acc}_{\leq 4}} \subsetneq Q_{\text{osc}}^{\text{SCIC}}, \\ \emptyset &\subsetneq Q_{\text{MinBG}} \subsetneq Q_{\text{osc}}^{\text{SCIC}}, \\ \emptyset &\subsetneq Q_{\text{Loc}_{3N}} \subsetneq Q_{\text{osc}}^{\text{SCIC}}, \\ \emptyset &\subsetneq Q_{\text{GlobProt}} \subsetneq Q_{\text{osc}}^{\text{SCIC}}, \end{aligned} \quad \text{with continuum image and complement in every target.}$$

The four subatlases are not identified with one another; they answer different target questions.

10 Explicit breaking, spontaneous breaking, and remnants

10.1 Neutrino matching, transparency, and spectator exclusion

Definition 4.3 distinguishes two logically independent requirements. A D5 breaker must lie on a target-bearing neutrino-matching path, and that same incidence must project to zero under the retained oscillation evaluators if the lift is to preserve the base packet. Physical coexistence on a common apex without target-bearing matching is insufficient.

Lemma 10.1 (Spectator-breaking exclusion). *Let Z be any finite source-coherent, physically valid $B - L$ -violating sector adjoined to a base source H . If Z has no source-active D5 neutrino-matching/bearer incidence on a current target-relative carrier path to H , then Z is stripped by PCR-3 spectator reduction and cannot establish $\text{ExpLift}(\bar{O}; H)$ for any operator/process carried only by Z .*

Proof. PCR-3 defines a target-inert spectator as a connected subdiagram with no current carrier-bearing incidence path to an instantiated target attachment or to the occupancy-bearing carrier core. Such a sector is removed before downstream survival is evaluated, and adjoining it does not alter the carrier trace. The explicit D5 target independently requires source support and neutrino matching. A decoupled breaker therefore lacks a required target record even when its internal field theory is physically consistent. D5-03 assigns it a cross/matching failure rather than a positive explicit-breaking lift. $\square$

Remark 10.2 (Sterile-singlet spectator as a negative control). Let N_s be an auxiliary Standard Model singlet of conventional rational $B - L$ charge +1 (primitive integral charge +3 on the marked faithful lattice) with Majorana process

$$\bar{O}_* = \frac{1}{2} N_s^T C N_s + \text{h.c.}, \quad |\Delta(B - L)_{\bar{O}_*}^{\text{conv}}| = 2, \quad |\delta_{\bar{O}_*}^{\text{int}}| = 6.$$

The finite local source block

$$\mathcal{L}_{*,c} = i N_s^\dagger \bar{\sigma}^\mu \partial_\mu N_s - \frac{1}{2} (c M_* N_s^T C N_s + \text{h.c.})$$

is physically valid for admitted nonzero c . If it is adjoined without a D5 neutrino-matching incidence to the B15 carrier graph, it remains a target-inert spectator. It must therefore be stripped and cannot count as a D5 explicit-breaking lift.

Proposition 10.3 (Conditional neutrino-matched transparent lift). *If $\text{NMTr}_{\text{D5}}(\bar{O}; H) = 1$, then*

there exists a one-apex D5 physical realization $W_{H,\bar{O}}$ with

$$\text{Res}_{\text{base}}(W_{H,\bar{O}}) \simeq W_H, \quad \Phi_4^{\text{ret}}(W_{H,\bar{O}}) = \Phi_4(W_H), \quad \Xi_{\text{osc}}^{\text{ret}}(W_{H,\bar{O}}) = \Xi_{\text{osc}}(W_H),$$

and $\bar{O}$ satisfies the frozen explicit-breaking target records on that same apex.

Proof. The predicate requires a source-active target-bearing neutrino-matching incidence and a common physical apex satisfying the D5 pure, cross, matching, continuation, and cone equations. Its final clause is precisely retained-oscillation transparency under the complete inherited evaluator and continuation grammar. D5-03 therefore gives physical survival, and D5-04 gives the base restriction and no-merger conclusion. $\square$

Remark 10.4 (Optional unitary representation after physical realization). If an already physical, neutrino-matched D5 process admits a finite contraction presentation A , the Julia–Halmos colligation

$$U(A) = \begin{pmatrix} (I - A^\dagger A)^{1/2} & -A^\dagger \\ A & (I - A A^\dagger)^{1/2} \end{pmatrix}$$

may be used as an exact representation of that process. It is subsequent to the physical process and matching theorem: it neither creates physicality, supplies the neutrino-matching incidence, nor defines a D5-standing response coordinate.

10.2 Explicit-breaking image

Definition 10.5 (Explicit-lift predicate). $\text{ExpLift}(\bar{O}; H)$ holds when there exists a physical D5 realization over H in which $\bar{O}$ is non-null and target-active, has $\delta_{\bar{O}} \neq 0$, possesses the fixed target-bearing neutrino-matching and continuation records, and restricts transparently to the oscillation target.

Set

$$Q_{\text{ExpLift}(\bar{O})} = \{q : \exists H \in \mathcal{N}_{\text{osc}}[q], \text{ExpLift}(\bar{O}; H)\}.$$

Theorem 10.6 (Explicit-breaking image).

$$\text{im } \rho_{\text{exp}[\bar{O}]} = Q_{\text{ExpLift}(\bar{O})}.$$

If a future or strengthened certificate proves $\bar{O} \in \bar{O}_{\text{D5}}^{\text{tr}}$, then conditionally

$$Q_{\text{ExpLift}(\bar{O})} = Q_{\text{osc}}^{\text{SCIC}}.$$

At the present frozen D5 certificate, $\text{Status}(\bar{O} \in \bar{O}_{\text{D5}}^{\text{tr}}) = \text{NotReady}$ for every operator type not already accompanied by a universal neutrino-matched transparent theorem. The exact current image is therefore the target-bearing intrinsic/matched support subatlas $Q_{\text{ExpLift}(\bar{O})}$; an unavailable or ill-typed operator yields the typed empty or update branch. Nonzero source activity alone does not determine the cardinality of an individual same-base lift fibre.

Proof. The first equality is the target acceptance condition after source completeness, simultaneous physical realization, target-bearing neutrino matching, and retained-base transparency. Proposition 10.3 proves the conditional full-base implication when the universal quantifier defining $\bar{O}_{\text{D5}}^{\text{tr}}$ is independently discharged. The present source and physical theorems do not establish that universal quantifier, and the spectator example shows why common-apex coexistence or a common charge label cannot do so. The full-base branch

therefore remains NotReady. Distinct nonzero activity magnitudes remain uncounted unless an admitted evaluator separates them. $\square$

Remark 10.7 (Intrinsic nonvacuity). The explicit-breaking target is nonvacuous. The marked law-only class q_W , for example, carries the source-active Weinberg operator as part of its neutrino mass and matching architecture, so $q_W \in Q_{\text{ExpLift}(\bar{O}_5)}$. This is an intrinsic target-bearing witness, not a universal attachment theorem.

10.3 Charge-normalization lock and covariantization by a breaking carrier

All divisibility, Smith-normal-form, and remnant calculations are performed in the primitive integral coordinate of the actual faithful charge lattice. Let q^{conv} denote the conventional rational $B - L$ normalization and let D_Λ be the least common denominator of the target-active faithful lattice. The transport to primitive integral coordinates is

$$\iota_\Lambda : q^{\text{conv}} \mapsto q^{\text{int}} = D_\Lambda q^{\text{conv}}, \quad \delta_O^{\text{int}} = D_\Lambda \Delta(B - L)_O^{\text{conv}}.$$

For the marked Standard-Model lattice, $D_\Lambda = 3$. Hence a source conventionally labelled $|\Delta(B - L)| = 2$ has $|\delta_O^{\text{int}}| = 6$, and a breaking scalar conventionally labelled $|B - L| = 2$ has $|q_\Phi^{\text{int}}| = 6$. A change of normalization is lawful only when the complete lattice, action, ineffective kernel, global quotient, operator charges, and breaking data are transported together.

Normalize the faithful $B - L$ charge lattice accordingly, following the violation-lattice and residual-symmetry discipline of SYM-7–SYM-9 [12, 13, 14]. In the remainder of this section, q_Φ and δ_O denote primitive integral coordinates unless a superscript “conv” is displayed. Let a breaking carrier Φ have nonzero integral charge q_Φ . An active source operator O with integral charge violation δ_O can be embedded in an invariant pre-breaking monomial

$$O \Phi^{n_+} (\Phi^\dagger)^{n_-}$$

exactly when

$$\delta_O + (n_+ - n_-)q_\Phi = 0$$

for some $n_+, n_- \in \mathbb{N}$. Equivalently,

$$q_\Phi \mid \delta_O$$

in the integral charge lattice.

Definition 10.8 (Spontaneous covariantization predicate). $\text{Cov}_\Phi(q_\Phi; H)$ holds when every source-active explicit $B - L$ violation in H lies in the ideal generated by q_Φ , the corresponding invariant pre-breaking source is finite and coherent, and a physical vacuum orbit with the required stabilizer and matching exists.

Theorem 10.9 (Spontaneous-breaking image).

$$\text{im } \rho_{\text{sp}[q_\Phi]} = Q_{\text{Cov}_\Phi(q_\Phi)}.$$

At the declared finite integral source scope, $q_\Phi = 1$ removes the arithmetic divisibility obstruction:

$$\delta_O \in 1\mathbb{Z} \quad \text{for every integral source violation } \delta_O.$$

The physical image nevertheless remains the exact $Q_{\text{Cov}_\Phi(1)}$ subatlas because the action, anomaly/global, vacuum, stabilizer, matching, and continuation clauses must still coexist on one apex. For $|q_\Phi| > 1$, the image is the corresponding divisibility-and-physicality subatlas. In the marked Standard-Model lattice, a conventionally charge-two breaking scalar and the marked Majorana/Weinberg violations all have primitive integral magnitude 6. Thus the

$|q_\Phi| = 6$ branch admits the marked Dirac, Majorana, and Weinberg support at the arithmetic level, while a general one-scalar branch admits the marked $|\Delta(B - L)^{\text{conv}}| = 2$ sources exactly when $|q_\Phi| \mid 6$. Additional breaking charges or a different faithful lattice require an explicit target update.

Proof. The spontaneous target requires an invariant pre-breaking source. For each finite active operator, invariance after adjoining powers of Φ is equivalent to the displayed integer equation. All operators can be covariantized simultaneously exactly when every violation lies in the generated ideal. D5-03 then attaches the breaking field, solves its vacuum orbit and stabilizer, and imposes matching so that the post-breaking realization retains the base packet. If $q_\Phi = 1$, every integral violation admits a finite covariantizing monomial; this proves only the arithmetic clause, not the remaining physical conditions. In the marked Standard-Model lattice, a source conventionally carrying $|\Delta(B - L)| = 2$ has $|\delta_O| = 6$, so one breaking charge satisfies the arithmetic condition precisely when $|q_\Phi|$ divides 6. $\square$

10.4 Several breaking charges and Smith normal form

For breaking carriers with charge matrix $Q_\Phi : \mathbb{Z}^r \rightarrow \Lambda$, the unbroken discrete quotient is computed from the cokernel after removing the continuous stabilizer and ineffective kernel. Smith normal form gives

$$UQ_\Phi V = \text{diag}(d_1, \dots, d_s, 0, \dots, 0), \quad d_i \mid d_{i+1},$$

and hence the finite remnant

$$\Gamma_{\text{SNF}} \cong \mathbb{Z}_{d_1} \times \dots \times \mathbb{Z}_{d_s},$$

modulo the faithful-action kernel and target-specific global identifications.

Definition 10.10 (Faithful remnant predicate). $\text{RemFaith}(\Gamma, q_\Phi; H)$ holds when H has a spontaneous lift with charge data q_Φ , the computed faithful stabilizer quotient is isomorphic to Γ , every retained operator has zero charge in the quotient, and every activated global-form, line, or defect record is compatible with that quotient.

Theorem 10.11 (Remnant image).

$$\text{im } \rho_{\text{rem}[\Gamma, q_\Phi]} = Q_{\text{RemFaith}(\Gamma, q_\Phi)}.$$

For a single faithful scalar of charge $n = |q_\Phi|$, the local remnant is $\mathbb{Z}_n$ modulo the common ineffective kernel, and an operator survives exactly when

$$\delta_O \equiv 0 \pmod{n}.$$

A target with a non-isomorphic declared Γ has empty fibre unless additional breaking or global-form data are supplied by an explicit target update.

Proof. The remnant is the subgroup fixing the vacuum, divided by elements acting trivially on the complete target-active source. The charge-lattice quotient or Smith normal form computes this object. An operator is invariant under the remnant exactly when its total charge vanishes in the quotient. Global-form and extended-sector data are then checked on the same physical apex. A declared remnant not isomorphic to the computed faithful quotient fails the target rather than being accepted by name. $\square$

Theorem 10.12 (D5-08 – Breaking, stabilizer, remnant, and matching classification). *The explicit, spontaneous, and remnant D5 fibres are exactly the groupoid-valued subatlases defined*

by $\text{ExpLift}(\mathcal{O})$, $\text{Cov}_\Phi(q_\Phi)$, and $\text{RemFaith}(\Gamma, q_\Phi)$. Every accepted remnant is reconstructed from one physical breaking diagram, its actual vacuum stabilizer, faithful quotient, operator congruence, activated global packet, and standing-preserving matching/continuation data. A remnant name, order, or chosen vacuum representative has no independent acceptance authority.

Proof. The explicit image theorem classifies target-bearing source-active violation support. The spontaneous image theorem adds the finite charge-ideal condition and the simultaneous action, anomaly/global, vacuum, stabilizer, matching, and continuation equations. Smith reduction computes the quotient of several breaking charges, after which the faithful-action kernel and target-specific global identifications are removed. The remnant image theorem then imposes the quotient selection rule and the global packet on the same apex. Together these results exhaust the source sorts and relations represented by D5-02; every remaining point lies in a failed, undefined, out-of-scope, or update stratum. $\square$

10.5 Marked-sector consequences

For the conventional marked source classes, transported to the primitive integral lattice:

- the Dirac class has $\delta_{\mathcal{O}}^{\text{int}} = 0$ and survives every charge-divisibility and remnant-congruence test compatible with its faithful action;
- the renormalizable Majorana and Weinberg classes have $|\Delta(B-L)^{\text{conv}}| = 2$, hence $|\delta_{\mathcal{O}}^{\text{int}}| = 6$ in the marked faithful lattice;
- a single scalar with $|q_\Phi^{\text{int}}| = 6$ leaves the raw local stabilizer $\mathbb{Z}_6$ before target-specific global identifications. If the declared global-form packet identifies an ineffective order-three subgroup, the faithful remnant is $\mathbb{Z}_2$; without that identification it remains $\mathbb{Z}_6$;
- in either of those two declared targets the marked $|\delta_{\mathcal{O}}^{\text{int}}| = 6$ Majorana/Weinberg support is trivial in the remnant, so neither the raw $\mathbb{Z}_6$ nor the quotient $\mathbb{Z}_2$ is an exact-Dirac selector;
- for a faithful one-scalar remnant $\mathbb{Z}_n$, the marked Majorana/Weinberg support is excluded exactly when $n \nmid 6$, subject to the complete global packet. The inequality $n > 2$ alone is not an exclusion criterion.

These are target-level selection rules, not terminal-carrier theorems.

10.6 Fibre geometry

A neutrino-matched transparent explicit-breaking theorem would give base-preserving lifts wherever its matching predicate holds, but no universal such family is proved here. Same-base cardinality is never inferred from the magnitude of a nonzero source-activity parameter. Spontaneous fibres may contain continuous vacuum or matching moduli only where those coordinates are independently target-standing. Remnant quotients add discrete and categorical moduli through faithful charge lattices, global forms, line sectors, and automorphism groups. The resulting fibres are therefore target-indexed moduli objects rather than finite model lists.

11 Target images and marginal classification

11.1 Subatlas notation

For an invariant source predicate P , write

$$Q_P = \{q \in Q_{\text{osc}}^{\text{SCIC}} : \exists H \in \mathcal{N}_{\text{osc}}[q] \text{ with } P(H)\}.$$

Existence is taken over the complete connected base groupoid and is therefore independent of chart choice and source provenance. The seven image formulas are

$\text{im } \rho_{\text{acc}}^{(4)} = Q_{\text{Acc}_{\leq 4}},$	$\text{im } \rho_{\text{bg}, \text{min}} = Q_{\text{MinBG}},$
$\text{im } \rho_{\text{loc}, 3N} = Q_{\text{Loc}_{3N}},$	$\text{im } \rho_{\text{glob}} = Q_{\text{GlobProt}},$
$\text{im } \rho_{\text{exp}[\mathcal{O}]} = Q_{\text{ExpLift}(\mathcal{O})},$	$\text{im } \rho_{\text{sp}[q_\Phi]} = Q_{\text{Cov}_\Phi(q_\Phi)},$
$\text{im } \rho_{\text{rem}[\Gamma, q_\Phi]} = Q_{\text{RemFaith}(\Gamma, q_\Phi)}.$	

11.2 Selection hierarchy

PCR-5C distinguishes the following outcomes:

- **full-base refinement:** $\text{im } \rho_\tau = Q_{\text{osc}}^{\text{SCIC}};$
- **proper-base selection:** $\emptyset \subsetneq \text{im } \rho_\tau \subsetneq Q_{\text{osc}}^{\text{SCIC}};$
- **base singleton:** $|\text{im } \rho_\tau| = 1;$
- **enriched singleton:** $|Q_{D5}(\tau)| = 1;$
- **empty target:** $\text{im } \rho_\tau = \emptyset;$
- **exact moduli:** nonempty fibres with finite, continuous, functional, categorical, or mixed geometry.

Excluding singletonhood is a target-specific theorem burden. It follows only when a proved base image or enriched fibre contains a non-singleton family. Neither an exact image formula nor the mere presence of continuous raw parameters suffices.

11.3 Target-wise classification

Target	Exact base image	Selection strength	Generic fibre geometry
$T_{\text{acc}}^{(4)}$	$Q_{\text{Acc}_{\leq 4}}$	Proper continuum image; continuum complement	Stable/categorical, with source-overlap moduli
$T_{\text{bg}, \text{min}}$	Q_{MinBG}	Proper continuum image; continuum complement	Stable/categorical background packet
$T_{\text{loc}, 3N}$	$Q_{\text{Loc}_{3N}}$	Proper continuum image; continuum complement	Categorical and mixed; no same-base coupling cardinality inferred
T_{glob}	Q_{GlobProt}	Proper continuum image; continuum complement	Categorical and mixed protector moduli

Target	Exact base image	Selection strength	Generic fibre geometry
$T_{\text{exp}}[\mathcal{O}]$	$Q_{\text{ExpLift}(\mathcal{O})}$	Exact target-bearing intrinsic or matched subatlas; universal full-base attachment remains NotReady	Mixed, empty, or update-dependent
$T_{\text{sp}}[q_\Phi]$	$Q_{\text{Cov}_\Phi(q_\Phi)}$	Exact divisibility plus one-apex physicality subatlas; $q_\Phi = 1$ removes only the arithmetic obstruction	Continuous and mixed vacuum/-matching moduli
$T_{\text{rem}}[\Gamma, q_\Phi]$	$Q_{\text{RemFaith}(\Gamma, q_\Phi)}$	Exact congruence and global-completion subatlas	Mixed discrete/-continuous/categorical

11.4 Established marginal cardinalities

Target or subdomain	Cardinality conclusion	Authority boundary
$T_{\text{acc}}^{(4)}, T_{\text{bg}, \text{min}}, T_{\text{loc}, 3N}, T_{\text{glob}}$	Non-singleton; continuum-strength	Each exact base image contains a continuum subatlas; neither the base image nor the enriched target quotient is singleton.
$T_{\text{exp}}[\mathcal{O}]$	NotReady	The exact image $Q_{\text{ExpLift}(\mathcal{O})}$ is known and particular operator targets have intrinsic witnesses, but no comparison-class-uniform non-singleton lower bound or universal full-base matched-transparent theorem is proved.
$T_{\text{sp}}[q_\Phi]$	NotReady	The exact divisibility-and-physicality image is known; no uniform non-singleton lower bound is proved for every fixed q_Φ .
$T_{\text{rem}}[\Gamma, q_\Phi]$	NotReady	The exact faithful-remnant image is known; no uniform non-singleton lower bound is proved for every fixed (Γ, q_Φ) .

11.5 The target family does not collapse

Two targets may admit the same marked base classes while retaining different enriched quotients and failure frontiers. For instance, in the marked primitive integral lattice, $T_{\text{sp}}[6]$ and a remnant target whose complete global-form packet reduces the raw $\mathbb{Z}_6$ stabilizer to a faith-

ful $\mathbb{Z}_2$ may both admit standard Dirac, Majorana, and Weinberg mass support. The spontaneous target nevertheless retains the continuous broken action and vacuum orbit, whereas the remnant target additionally fixes the faithful quotient and its selection rules. Their target packets are therefore inequivalent.

Theorem 11.1 (D5-09 – Exact base-image, selection, and row-specific cardinality theorem). *The endpoint matrix above is exact at connected SCIC resolution. Every nonempty image is computed from the complete lift atlas; every excluded base class carries a first-separator certificate; and no target fibre is merged with another. Cardinality conclusions are exactly those listed in the proved marginal-cardinality table. The overall PCR-5D5 endpoint is therefore an exact target-indexed mixed selection object, not a universal carrier count.*

Proof. D5-05 identifies each image with the set of nonempty connected fibres. Sections 9 and 10 prove that fibre nonemptiness is equivalent to the displayed invariant source predicates. The properness and continuum-strength statements for the first four targets follow from the accepted and excluded continuum witnesses in Appendix C. For explicit breaking, Section 10 proves the exact target-bearing image and the spectator-exclusion boundary. A universal neutrino-matched transparent attachment would imply a full base image, but no such universal theorem is established here, so the explicit-breaking target retains NotReady cardinality status. The spontaneous and remnant targets follow from the charge-ideal, simultaneous-physicality, and faithful-quotient theorems, which determine exact images without supplying a uniform lower cardinality for every fixed target. D5-04 gives no merger, and D5-01 gives target separation. $\square$

11.6 Consequences for role occupancy

The continuum base images of the first four targets exclude marginal enriched singletons. For the explicit, spontaneous, and remnant targets, D5 makes no singleton or non-singleton assertion beyond the separately proved cardinality status. In particular, a decoupled $B - L$ -violating spectator cannot be used to create a full-base explicit branch; universal neutrino-matched transparency remains NotReady. For every target, marginal survival is distinct from terminal physical realization. D5 contains neither the common D1–D5 source nor the terminal eligibility, compatibility, and Role-Occupancy Closure calculation required for positive carrier identity. The joint target must therefore be evaluated on one common physical source rather than assembled from selected marginal outcomes.

D5 is neither a universal independent fibre coordinate of the D1/D2 type nor a single proper selector of the D3-successor type. It is a target-indexed family. Accidental, minimal-background, local, and exact-global targets select different proper subatlases; explicit, spontaneous, and remnant targets have exact operator- and charge-dependent images. Symmetry and charge have carrier-selective force only after their target type is specified, and both cardinality and D3-comparison conclusions remain target-specific.

12 Symmetry bearers and spectral sectors

12.1 Bearer-support hypergraph

For an accepted D5 point Y , let $\mathfrak{B}_{D5}(Y)$ be the finite bearer hypergraph. Its vertices comprise:

1. the common continuation-bearing carrier reference inherited from the B15 standing

core;

2. standing spectral-sector references $P_i N$, treated as subobjects rather than terminal nodes;
3. source blocks such as singlet sectors, breaking carriers, anomaly-cancelling family packets, and global-form objects;
4. irreducible relational or higher-arity bearer nodes.

The hyperedges are the target-active action, charge, protector, violation, breaking, matching, anomaly, and continuation incidences.

12.2 No-repair source composition

Let $K(Y)$ be the finite set of active source kinds. The no-repair sufficiency map

$$N_{\tau, Y} : 2^{K(Y)} \rightarrow \{0, 1, \perp\}$$

asks whether a retained source subset continues to realize the complete D5 target after its complement is deleted without retuning. For an accepted point, define the nonempty antichain of inclusion-minimal sufficient subsets

$$\text{MinSupp}_{\tau}(Y) = \min_{\subseteq} \{A \subseteq K(Y) : N_{\tau, Y}(A) = 1\}.$$

The reported S/R/P/Y class is assigned by the following disjoint precedence rule:

- $Y \iff \text{MinSupp}_{\tau}(Y) = \{K(Y)\},$
- $S \iff |\text{MinSupp}_{\tau}(Y)| = 1 \text{ and its unique member is a proper subset of } K(Y),$
- $P \iff |\text{MinSupp}_{\tau}(Y)| \geq 2 \text{ and every } A \in \text{MinSupp}_{\tau}(Y) \text{ has } |A| > 1,$
- $R \iff |\text{MinSupp}_{\tau}(Y)| \geq 2 \text{ and at least one } A \in \text{MinSupp}_{\tau}(Y) \text{ is a singleton.}$

Thus “single-route” means one unique minimal sufficient route, not necessarily one source kind. The separate Y case records the stronger fact that the complete active source set itself is uniquely minimal.

12.3 Spectral-incidence graph and covariance

Let a target-complete neutrino propagation carrier have standing spectral resolution

$$M_{\nu} = \sum_{i \in I} m_i P_i.$$

Define the graph $\mathcal{G}_{\text{spec}}(Y)$ on the vertex set I by placing an edge $i \sim j$ whenever a retained oscillation-boundary, matching, or continuation incidence has nonzero cross support between $P_i N$ and $P_j N$. Equivalently, an edge is present when a target-admitted operator X satisfies

$$P_i X P_j \neq 0.$$

The graph is determined by the complete retained source and standing trace, not by a chosen PMNS chart.

Definition 12.1 (Exact spectral-covariance condition). $\text{ExactCov}_{D5}(Y)$ holds when the fixed D5 target supplies an exact continuous action or an exact residual action on the complete target-relevant neutrino mass response and retained spectral-incidence structure, and every

target-active incidence defining $\mathcal{G}_{\text{spec}}(Y)$ is covariant under that action. Bounded accidental invariance in a restricted operator envelope does not imply $\text{ExactCov}_{\text{D5}}(Y)$ when the complete mass response contains an active $B - L$ -violating operator.

Theorem 12.2 (Connected spectral charge constancy). *Suppose $\text{ExactCov}_{\text{D5}}(Y)$ holds for an exact continuous action generated by Q , with*

$$QP_i = q_i P_i.$$

Then $q_i = q_j$ whenever i and j lie in the same connected component of $\mathcal{G}_{\text{spec}}(Y)$.

Proof. For an edge $i \sim j$, choose an admitted incidence X with $P_i X P_j \neq 0$. Exact covariance gives $[Q, X] = 0$ on the relevant interface. Multiplying by P_i and P_j yields

$$0 = P_i [Q, X] P_j = (q_i - q_j) P_i X P_j.$$

The final factor is nonzero, so $q_i = q_j$. Equality then propagates along paths in the graph. $\square$

Theorem 12.3 (Residual-character constancy on connected spectral components). *Suppose $\text{ExactCov}_{\text{D5}}(Y)$ holds for an exact residual group Γ , and let χ_i be the residual character on $P_i N$. Then $\chi_i = \chi_j$ whenever i and j lie in the same connected component of $\mathcal{G}_{\text{spec}}(Y)$.*

Proof. For an edge $i \sim j$, exact residual covariance of a nonzero incidence $P_i X P_j$ implies

$$\chi_i(g) P_i X P_j = \chi_j(g) P_i X P_j \quad \forall g \in \Gamma.$$

Nonzero cross support gives $\chi_i(g) = \chi_j(g)$ for every g , and equality propagates along connected paths. $\square$

Corollary 12.4 (Connected spectral charge/character uniformity and nonpromotion). *Whenever $\text{ExactCov}_{\text{D5}}(Y)$ holds, exact continuous charge or exact residual character is uniform on each connected component of $\mathcal{G}_{\text{spec}}(Y)$. In the connected continuous case,*

$$Q_{B-L}|_N = q_N \mathbf{1}_N.$$

Thus a nondegenerate mass spectrum does not, by itself, produce several symmetry-distinguished terminal-carrier identities.

Nonclaim 12.5 (Uniformity is not positive terminal identity). The preceding results concern exact symmetry action on the retained spectral-incidence object. They do not prove that graph connectedness is sufficient for one terminal carrier, identify the spectral graph with the terminal SCIC quotient, or discharge terminal eligibility, compatibility, and Role-Occupancy Closure. The status

ConnectedSpectralChargeUniformity

is assigned only when the exact continuous or residual covariance condition is satisfied.

12.4 Bearer classification

The reported bearer status is determined by the following precedence relation.

1. If the target/source data leave bearer status incomplete, return `BearerNotReady`.
2. If the target separator requires an irreducible higher-arity source packet, return `RelationalHigherAritySymmetryStanding`.

3. If distinct spectral components have independently complete charge, protector, failure, and continuation signatures, return `SectorwiseCompleteProtectorSignature`.
4. If an exact action distinguishes spectral components but the complete protector signature is absent, return `SectorwiseSpectralSymmetryStanding`.
5. If $\text{ExactCov}_{D5}(Y)$ holds and none of the preceding sectorwise or higher-arity cases takes priority, return `ConnectedSpectralChargeUniformity`.
6. Otherwise return `BearerNotReady`.

The underlying predicates may overlap. The precedence relation selects one reported status without asserting the absence of lower-priority structure.

Theorem 12.6 (D5-10 – Bearer-resolution, conditional spectral charge-uniformity, and source-composition closure). *Every accepted D5 class receives one total bearer status, one exact S/R/P/Y profile, one minimal standing separator, and one terminal-promotion status in*

$\{\text{NotActivated}, \text{Candidate}, \text{Deferred}\}.$

No D5 point receives the status “independent terminal carrier.” Connected spectral charge uniformity is available only under

$\text{ExactCov}_{D5}(Y).$

In particular, if the complete mass response contains an active $B - L$ violation outside the bounded accidental envelope, then

$T_{\text{acc}}^{(4)} \not\Rightarrow \text{ConnectedSpectralChargeUniformity}.$

Proof. The active source hypergraph is finite, and acceptance makes the full active set sufficient, so the minimal-support antichain is nonempty. Exactly four cases remain. If the antichain has one member and that member is the full active set, the profile is Y; if it has one proper member, the profile is S; if it has at least two members and all have cardinality greater than one, the profile is P; and if it has at least two members with at least one singleton, the profile is R. These cases are exhaustive and pairwise disjoint. The bearer predicates are determined by the complete source trace and may overlap, while the precedence relation assigns a unique reported bearer status. Irreducible higher-arity support has relational priority. Otherwise, complete protector signatures and exact spectral-action data determine the sectorwise cases. Connected spectral uniformity follows only under the exact continuous or residual covariance condition. In its absence, bearer resolution fails closed rather than importing a symmetry from a bounded operator envelope. Terminal eligibility, joint-target survival, compatible role occupancy, and the final carrier quotient are not coordinates of the D5 target, so terminal promotion lies outside the codomain. $\square$

12.5 Target-wise bearer diagnoses

Target	Generic bearer status	Reason
$T_{\text{acc}}^{(4)}$	Bearer not ready at bounded accidental scope	The $d \leq 4$ envelope may preserve $B - L$ while the complete neutrino mass response contains an active higher-dimensional violation, such as a Weinberg source. Bounded accidental standing therefore does not authorize exact spectral charge uniformity.

Target	Generic bearer status	Reason
$T_{bg,min}$	Relational/higher-arity	The anomaly coefficient is an irreducible sum over the complete chiral spectrum and the background probe.
$T_{loc,3N}$	Relational/higher-arity; conditional continuous uniformity	Cancellation requires the complete family-indexed chiral packet. If the accepted point also satisfies exact local covariance on the complete retained spectral-incidence structure, connected charge uniformity follows as a secondary consequence.
T_{glob}	Connected spectral charge uniformity	The exact-global target supplies the continuous covariance required by ExactCov_{D5} on every target-active mass and continuation incidence.
$T_{exp}[\mathcal{O}]$	Usually relational/higher-arity	Breaking is borne by an operator or process incidence involving all fields or states in $\mathcal{O}$; explicit violation does not furnish an exact continuous charge action.
$T_{sp}[q_\Phi]$	Relational/higher-arity	The action, breaking carrier, vacuum orbit, matching, and continuation are jointly necessary.
$T_{rem}[\Gamma, q_\Phi]$	Relational/higher-arity; conditional residual- character uniformity	The remnant is the faithful quotient of the complete breaking and global data. If its exact residual action is established on the complete retained spectral incidence, residual characters are componentwise uniform.

12.6 Consequence for the one-carrier hypothesis

Standing Carrier Identity Closure and Role-Occupancy Closure [10, 11] prevent D5 from proving the positive terminal claim that the neutrino is one carrier with three spectral sectors. D5 proves the narrower conditional nonpromotion result needed before such a theorem can be attempted:

$$\begin{aligned} &\text{exact continuous or residual covariance} + \text{connected retained spectral incidence} \\ &\implies \text{uniform symmetry charge or character on that component.} \end{aligned}$$

Mass-spectral multiplicity therefore does not, by itself, generate symmetry-distinct terminal carriers. The bounded accidental target does not supply this conclusion when the complete mass response violates $B - L$. Sectorwise symmetry standing may occur only where the retained exact-action structure supports it, and terminal promotion remains a candidate or deferred status.

13 Comparison with the D3 active-closure coordinate

13.1 Comparison scope

Only the successor active-closure target and its publication theorem are used here [8]:

$$T_{D3}^{1,3} = \text{THUCL-D3-1.3.}$$

Its retained quantitative separator is the universal process-tensor contraction

$$S_{D3}(Y) = C_{ee}^{\text{op},1.3}(Y) = W_Y * T_{e,\text{act}}^{U,\text{ret}},$$

with threshold and coverage semantics fixed by the 1.3 target. The distinct THUCL-D3-1.1 Global Fibre Completion programme is neither completed nor revised here. A statement carrying the 1.3 tag is not a theorem at the 1.1 target.

For a D5 target τ , let

$$S_{D5,\tau}(Y)$$

denote the minimal separator obtained in Sections 9–12. It may be an operator-support predicate, anomaly vector, protector relation, violation-lattice class, stabilizer quotient, or faithful-remnant congruence. Even if $\text{im } \rho_{D3}^{1.3}$ and $\text{im } \rho_{D5,\tau}$ were equal, their equality would not identify these separator objects.

Definition 13.1 (Marginal functional dependence). On a common physical comparison subcategory $\mathcal{C}$, S_1 is functionally dependent on S_2 when a target-preserving map f exists such that

$$S_1 = f \circ S_2$$

on every object and arrow of $\mathcal{C}$. The coordinates are marginally independent when neither factors through the other and each admits a nonconstant physical deformation with the other held fixed. If either target-faithful deformation direction is unavailable, the comparison returns NotReady rather than inferring dependence or independence.

13.2 A $B - L$ -preserving active-closure deformation

Begin with a full-rank Dirac three-singlet realization carrying exact $B - L$ and a closed active process. Adjoin a neutral hidden propagation sector with the same $B - L$ charge as the active neutrino carrier and couple it through a $B - L$ -covariant unitary dilation. Let $\delta \in [0, \delta_*]$ be the retained active-to-hidden leakage probability. The deformation preserves the D5 exact-global action, charge lattice, anomaly record, and protector, while the retained D3 active normalization becomes

$$C_{ee}^{\text{op},1.3}(Y_\delta) = 1 - \delta.$$

The family may be chosen with $\delta_* > 0.013$, so that it contains both 1.3-pass and 1.3-fail points.

Lemma 13.2 (D3 variation at fixed D5 exact-global standing). *There exists a nondegenerate interval I_3 and a physical family Y_δ , $\delta \in I_3$, such that*

$$S_{D5,\text{glob}}(Y_\delta) \equiv \text{ExactGlobalPass}, \quad S_{D3}(Y_\delta) = 1 - \delta.$$

In particular, S_{D3} does not factor through $S_{D5,\text{glob}}$.

Proof. Assign the hidden sector the same faithful $B - L$ character as the connected active neutrino sector. Every mixing, propagation, projection, and continuation arrow is then $B - L$ covariant, and the global protector remains valid because no active $\Delta(B - L) \neq 0$ source is introduced. The dilation is unitary on the active-plus-hidden space. The retained D3 evaluator coarsens to the active charged-current outcomes and omits the hidden outcome, giving $1 - \delta$. Since the D5 exact-global packet is constant while the D3 scalar varies, the latter cannot be a function of the former on this family. $\square$

Corollary 13.3 (D3 variation at fixed accidental $d \leq 4$ standing). *The same family satisfies*

$$S_{D5,acc}(Y_\delta) \equiv \text{AccidentalPass}_{d \leq 4}.$$

Hence S_{D3} does not factor through $S_{D5,acc}$.

Proof. Exact-global covariance preserves every target-active operator, and therefore every operator in the bounded $d \leq 4$ envelope. The hidden dilation adds only $B - L$ -covariant relations and leaves the accidental target unchanged. $\square$

13.3 Closure-preserving $B - L$ deformations

Choose a three-active-state unitary propagation family with no hidden leakage and

$$C_{ee}^{op,1.3} = 1.$$

In this subsection, the displayed label $\Delta(B - L) = 2$ is the conventional rational coordinate; on the marked faithful lattice it corresponds to primitive integral magnitude $|\delta^{\text{int}}| = 6$. Add such a source-active deformation on a D5-transparent branch: a Weinberg-type source law, a renormalizable singlet-Majorana source, or another target-indexed operator whose retained active process is held fixed along the base-frame family. Let c be its source-activity parameter; only the null/non-null boundary is used. The D3 active-closure record remains constant while the relevant D5 operator-support or protector separator changes at that boundary.

Lemma 13.4 (Accidental and exact-global variation at fixed D3 active closure). *There exists a physical family Z_c , with c ranging over a source domain containing 0 in its closure and at least one active value, such that*

$$S_{D3}(Z_c) \equiv 1,$$

while both $S_{D5,acc}(Z_c)$ and $S_{D5,glob}(Z_c)$ are nonconstant for suitable source-authorized $\Delta(B - L) = 2$ choices. Consequently neither D5 separator factors through S_{D3} .

Proof. For the accidental target, choose a renormalizable singlet-Majorana source. At $c = 0$ the incidence is source-null and removed by activity reduction; for $c \neq 0$ it is an active dimension-three violation. For the exact-global target, either the same source or a Weinberg source changes protector status. In both cases the retained propagation channel remains a normalized active unitary process. The D3 evaluator is therefore identically one while the D5 separator changes. $\square$

13.4 Targets whose comparison remains unresolved

The explicit-breaking target requires a source-active neutrino-matching/bearer incidence on the actual base carrier architecture. A decoupled $B - L$ -violating product is a spectator and cannot supply this record. The stated comparison class contains no universal family that both varies D3 closure while preserving the complete explicit target and varies the explicit separator while preserving D3 closure. Intrinsic explicit operators may be target-bearing, but changing their activity may also alter the base neutrino mass or matching architecture. The generic explicit-breaking comparison is therefore NotReady.

The other unresolved targets require additional records not preserved by the available deformation pairs:

- $T_{bg,min}$ fixes the minimal chiral inventory, which the hidden-sector D3 deformation changes.

- $T_{\text{loc},3N}$ requires the complete faithful local action, covariant support, and local anomaly packet; preservation under a D3-varying family has not been proved.
- $T_{\text{sp}}[q_\Phi]$ requires one fixed action, breaking carrier, vacuum orbit, stabilizer, matching, and continuation packet in both directions.
- $T_{\text{rem}}[\Gamma, q_\Phi]$ additionally requires preservation of the faithful quotient, operator congruences, and activated global packet.

The absence of these deformation pairs implies neither dependence nor independence.

13.5 Target-indexed comparison theorem

Theorem 13.5 (D5-11 – Target-indexed D3–D5 comparison). *At the target-qualified D3 scope THUCL-D3-1.3,*

$$\text{D3D5Status}(1.3, T_{\text{acc}}^{(4)}) = \text{D3D5Status}(1.3, T_{\text{glob}}) = \text{IndependentCoordinate}.$$

The minimal-background, local-3N, explicit-breaking, spontaneous, and remnant rows return NotReady. No marginal common-generator factorization is established for any row.

Proof. The accidental and exact-global rows follow from the paired deformation lemmas: D3 varies at fixed D5 standing, and the corresponding D5 separator varies at fixed D3 closure. For every other target, at least one target-faithful deformation direction is unavailable. In particular, a physically valid explicit-breaking spectator is not a target-bearing D5 lift, and no universal neutrino-matched transparent deformation pair is established. The definition therefore requires NotReady. Such a status is evidence of neither dependence nor independence. None of the marginal constructions provides a standing-bearing minimal common generator on a simultaneous D3–D5 category. $\square$

Corollary 13.6 (No image-based common generator). *Equal, nested, or overlapping D3 and D5 base images cannot establish a common generator. Such a theorem would require a target-authorized invariant G and factorization maps*

$$S_{\text{D3}} = f_3(G), \quad S_{\text{D5},\tau} = f_5(G)$$

on one simultaneous physical category, together with a proof that G is standing-bearing and minimal at the joint target.

13.6 Interpretation

The decided targets show that active-closure standing and the accidental and exact-global symmetry coordinates are not alternative names for one marginal separator. The explicit-breaking comparison remains unresolved once target-bearing neutrino matching is imposed, and the other unresolved targets remain open because their complete targets require stronger inventory, gauge, vacuum, stabilizer, quotient, or matching data. PCR-5E must retain both the positive independence results and every NotReady boundary. Dependence, common generation, or an emergent higher-arity relation can be established only on the simultaneous physical category of the joint target.

14 Marginal data for the joint neutrino target

14.1 The D5 marginal structure

PCR-5E must evaluate D5 on the same joint source as the other phenomenon targets. A list of surviving base classes is therefore insufficient. Define

$$\mathfrak{H}_{D5} := \left(\mathcal{F}_{D5}, \right. \\ \left. \{ \mathbb{T}_\tau, \mathcal{N}_{D5}(\tau), R_\tau, \rho_\tau, \mathcal{C}_\tau, Q_\tau, S_{D5,\tau}, \mathfrak{B}_\tau, \right. \\ \left. \text{MinSupp}_\tau, \partial\text{Fail}_\tau, \text{CardStat}_\tau, \text{D3D5Stat}_\tau, \text{TermProm}_\tau \}_{\tau \in \mathcal{F}_{D5}}, \right. \\ \left. \mathfrak{C}_{3,5}, \text{Prov}_{D5} \right).$$

Here $\mathfrak{C}_{3,5}$ is the target-indexed D3–D5 comparison object, and Prov_{D5} records source identities, overlap maps, theorem and target versions, and the hypotheses governing each conclusion.

Definition 14.1 (Sufficiency for joint reconstruction). The D5 marginal object is sufficient for joint reconstruction when the joint analysis can, without reinterpretation:

1. recover every target fingerprint and applicability rule;
2. identify the same B15/PCR-3 source nodes inside a proposed joint source;
3. reconstruct every marginal physical apex, restriction map, connected fibre, and overlap;
4. evaluate the D5 separator, bearer, source-composition, and failure objects;
5. recover the target-specific cardinality and D3-comparison conclusions, including every explicit NotReady boundary;
6. distinguish marginal pass, marginal fail, undefined, out-of-scope, source-update, and target-update cases;
7. preserve the prohibition on terminal promotion within D5.

14.2 Components required by the joint analysis

For each target τ , the marginal object contains:

Component	Mathematical content
Target	The member of $\mathcal{F}_{D5}$, its parent, complete record fingerprint, applicability data, and update boundary.
Source comparison class	The SOE–D5–1 normal-form scope, exact reduct to the B15 base, activity rule, and source groupoid.
Physical realization	The one-apex residual system, transparent locus, acceptance condition, and common-cone data.
Restriction	The concrete R_τ , descended ρ_τ , no-merger theorem, and transport along base arrows.
Fibres	Framewise solution objects, connected homotopy fibres, codescent maps, isotropy, and coarse geometry.
Base image	The invariant source predicate, distinguished witnesses, inclusion and noninclusion relations, and selection strength.
Cardinality	The target-specific cardinality theorem, its exact scope, or an explicit NotReady where no uniform bound is known.
Separator	The minimal target separator or separating family rather than a family name or fitted parameter.

Component	Mathematical content
Bearer	The bearer status, spectral-incidence components, the ConnectedSpectralChargeUniformity interpretation boundary, and terminal-promotion status.
Source composition	The active source-kind set, no-repair deletion map, minimal sufficient subsets, and S/R/P/Y class.
Failure locus	The complete fail, undefined, ROOS, source-update, and target-update complement with first-separator provenance.
Overlaps	Pairwise and higher target/source overlaps with target tags retained.
D3 comparison	The 1.3 target tag, target-specific status, proof record or NotReady reason, every established subdomain, and the prohibition on transport to 1.1.

14.3 Target-indexed D3 comparison

The marginal comparison is:

D5 target	D3 status	Established boundary
$T_{\text{acc}}^{(4)}$	IndependentCoordinate	Two target-faithful deformation directions are proved.
$T_{\text{bg},\text{min}}$	NotReady	No D3-varying family preserves the fixed minimal chiral inventory.
$T_{\text{loc},3N}$	NotReady	Preservation of the complete local action and anomaly packet is not proved in both directions.
T_{glob}	IndependentCoordinate	Two target-faithful deformation directions are proved.
$T_{\text{exp}}[\mathcal{O}]$	NotReady	No universal neutrino-matched transparent two-direction deformation pair is proved; decoupled breakers are spectators.
$T_{\text{sp}}[q_\Phi]$	NotReady	The action, vacuum, stabilizer, matching, and continuation packet are not preserved in a proved pair.
$T_{\text{rem}}[\Gamma, q_\Phi]$	NotReady	The faithful quotient, congruence, global packet, and matching data are not preserved in a proved pair.

14.4 Why marginal survivor sets are insufficient

The joint target cannot replace $\mathfrak{H}_{\text{D5}}$ by

$$\bigcup_{\tau \in \mathcal{F}_{\text{D5}}} \text{im } \rho_\tau, \quad \bigcap_{\tau \in \mathcal{F}_{\text{D5}}} \text{im } \rho_\tau, \quad \text{or} \quad \prod_{\tau \in \mathcal{F}_{\text{D5}}} \pi_0 \mathcal{L}_\tau(q).$$

The union discards target identity, the intersection discards simultaneous compatibility, and the product permits marginal lifts drawn from different sources, vacua, global forms, or matching maps. PCR-5C instead requires one source-coherent enriched point. Likewise, a

NotReady target cannot be replaced by an assumed dependence, independence, or common-generator relation.

Theorem 14.2 (D5-12 – Completeness of the marginal D5 interface). *The object $\mathfrak{S}_{D5}$ is reconstruction-sufficient. Every field required for the PCR-5E marginal D5 leg is either total or carries an explicit NotReady, source-update, or target-update certificate. Lawful D5 presentation changes induce isomorphic marginal interface objects, and no downstream characterization may alter completed D5 fibre membership or erase a row-specific authority boundary.*

Proof. D5-01 fixes the target fingerprints; D5-02 and the source-representation theorem supply source identities and raw complements; D5-03 establishes simultaneous physical realization; D5-04 establishes retained restriction and no merger; D5-05 establishes connected fibres and codescent; D5-06–D5-09 determine the source-semantic images, separators, and target-specific cardinality conclusions; D5-10 determines bearer and source-composition data; and D5-11 establishes the target-indexed D3 comparison, including its explicit NotReady boundaries. Each component is functorial under lawful arrows and carries framed provenance. Together these results determine the complete D5 marginal structure. PCR-5C no-characterization-backflow makes completed marginal membership final at its target scope while preserving unresolved fields as unresolved. $\square$

14.5 Standing-coordinate incidence

The D5 contribution to the phenomenon-to-separator incidence relation is

$$I_{D5} = \{(\tau, s) : \tau \in \mathcal{F}_{D5}, s \text{ is a minimal D5 standing separator for } \tau\}.$$

The target tag is part of the ordered pair. An anomaly obstruction, exact protector, violation ideal, and remnant congruence cannot be collapsed into one generic “symmetry coordinate.” A joint theorem may show that several targets factor through one invariant, but that would be a conclusion of the simultaneous calculation.

The target-indexed D5 objects are fixed marginal inputs to the one-point PCR-5E realization problem. No D5 bearer status may be treated as terminal identity; no common generator may be inferred from marginal image alignment; no NotReady D3-comparison target may be replaced by a decided relation; and no D5 failure or update frontier may be erased.

15 Robustness, falsifiability, and scope

15.1 Load-bearing theorem chain

The endpoint depends on the ordered implications

$$\begin{aligned} D5-01 &\Rightarrow D5-02 \Rightarrow D5-03 \Rightarrow D5-04 \Rightarrow D5-05 \\ &\Rightarrow D5-06/07/08 \Rightarrow D5-09 \Rightarrow D5-10/11 \Rightarrow D5-12. \end{aligned}$$

A counterexample must preserve the target and comparison class and invalidate a named implication. Preference for a particular model, enlargement of the target, or a change of

ontological vocabulary does not by itself contradict the theorem chain.

15.2 Counterexample requirements

Proposed objection	Same-scope defeating construction required	Result affected
Target collapse	Exhibit two target fingerprints in $\mathcal{F}_{D5}$ that agree on every required record, applicability rule, acceptance predicate, continuation, and failure boundary.	D5-01.
Source omission	Give a finite same-target D5 symmetry/charge architecture not representable by the SOE–D5–1 source sorts, operations, relations, and finite hybrid closure.	SLR-D5-1 and D5-02.
Failure of simultaneous realization	Give an accepted point whose anomaly, vacuum, matching, action, or global records cannot coexist on one physical apex.	D5-03 and the affected fibre.
Failure of transparency	Give a D5-only cell with a genuine old-target output on a point classified transparent, or a licensed old continuation not preserved by restriction.	D5-04 and every downstream fibre using that point.
Spectator promotion	Adjoin a physically valid $B - L$ -violating sector with no source-active target-bearing neutrino matching to the retained carrier architecture and nevertheless classify it as an explicit-breaking lift.	D5-03/D5-08; the correct disposition is PCR-3 spectator stripping and no D5 explicit-breaking lift.
Fibre omission	Give an accepted source-faithful physical lift absent from the codescent atlas, or two counted components joined by a lawful overlap.	D5-05.
Failure of charge descent	Give a lawful lattice, basis, or family transformation that changes target pass/fail, or two nonisomorphic faithful actions erased by the quotient.	D5-06.
Partial anomaly substitution	Preserve the target while omitting an active anomaly coefficient and still produce a complete target record.	D5-06/D5-07.
Local-to-global promotion	Construct the required global action, quotient, line/defect, continuation, and global-obstruction packet from local polynomial cancellation alone at the same target.	D5-07.

Proposed objection	Same-scope defeating construction required	Result affected
Remnant by label	Produce a faithful remnant satisfying the target without a breaking carrier, vacuum stabilizer, charge-lattice quotient, operator rules, and global qualification.	D5-08.
Normalization mismatch	Perform divisibility or remnant arithmetic after changing displayed charge normalization without transporting the complete faithful lattice, ineffective kernel, and global quotient.	D5-06/D5-08; the result must be rejected as ill-typed.
Source-composition gap	Give an accepted finite source whose unique minimal sufficient route is a proper multi-kind subset of the active source set and show that it receives no S/R/P/Y value.	D5-10; the repaired precedence assigns S.
Dirac-continuum source failure	Reproduce a varying post-EWSB block matrix while failing to supply gauge-invariant pre-EWSB operators, anomaly-neutral vectorlike support, one simultaneous physical apex, or a retained threshold/matching separator over a nondegenerate interval.	Theorem 9.11, D5-07, and the local/global accepted-side continuum in D5-09.
Exact-response failure	Hold only a Schur-complement coefficient fixed while the exact light pole spectrum, canonical charged-current normalization, event-complete response, or heavy applicability status varies.	Rejected by the exact unitary factorization and response calculation in Theorem 9.11.
Matching-basis incompleteness	Delete one independent exact- $B - L$ light source operator while retaining the corresponding independent B15 coefficient, or make the source-to-matched coefficient map non-surjective after the Ward/EOM/field-redescription quotient.	Theorem 9.12 must return <code>NotReady</code> MatchingBasisIncomplete rather than <code>Pass</code> .
Continuation-stratum crossing	Move the interval so that a compensating source coefficient changes activity class, a heavy threshold enters a target-facing continuation range, or the benchmark reaches an applicability/failure frontier.	Theorem 9.12 must return <code>NotReady</code> ContinuationStratumCrossing rather than treating a discrete status as continuously transportable.
Base-image counterexample	Give $q \in Q_{\text{osc}}^{\text{SCIC}}$ satisfying the stated source predicate but with empty fibre, or failing the predicate while admitting an accepted physical lift.	D5-09 for that target.

Proposed objection	Same-scope defeating construction required	Result affected
Bearer counterexample	On a point certified by ExactCov_{D5} , give a connected spectral component carrying nonuniform exact charge or residual character; or show that $\text{ConnectedSpectralChargeUniformity}$ was assigned without exact covariance or promoted to terminal identity.	D5-10 and the source-representation bearer clause.
Terminal-promotion counterexample	At the same marginal target, provide the complete continuation signature, terminal eligibility and compatibility, and a unique ROC orbit.	The terminal-promotion boundary; such a result would be a successor theorem.
D3–D5 comparison	Refute an established target-faithful deformation pair, provide the missing two-direction construction for a target marked NotReady , or give an exact common invariant with factorization maps on one simultaneous physical category.	D5-11.
Insufficient joint input	Give a D5 marginal object required by PCR-5E that cannot be reconstructed from $\tilde{\mathfrak{H}}_{D5}$.	D5-12.

15.3 Direct falsifiers

The following findings would revise the stated endpoint:

1. a lawful B15 base class satisfying $\text{Acc}_{\leq 4}$, MinBG , Loc_{3N} , or GlobProt but lacking the corresponding physical lift;
2. a lawful physical lift of one of those targets whose base class does not satisfy the stated predicate;
3. a connected spectral-incidence graph carrying two different exact continuous $B - L$ charges on vertices joined by a nonzero covariant incidence;
4. a remnant target accepted although its declared Γ is not the faithful vacuum-stabilizer quotient;
5. a lawful charge normalization that changes the target survivor set;
6. a marked divisibility or remnant calculation that treats conventional charge 2 as primitive integral charge 2, or infers $\mathbb{Z}_2$ without the required global quotient;
7. an accepted finite source whose minimal-support antichain is not assigned exactly one S/R/P/Y value by the stated precedence rule;
8. failure of the source-authorized vectorlike Dirac family to define gauge- and $B - L$ -invariant one-apex PCR-4/B15 survivors throughout I_{RG} , incompleteness of the independent light matching basis, crossing of an activity/applicability/failure frontier, failure of the explicit finite-order matching right inverse to keep every target-retained light continuation in one accepted packet, or failure of M^{match} to separate distinct heavy thresholds;

9. for $T_{\text{acc}}^{(4)}$ or T_{glob} , a D3–D5 comparison point showing that an asserted independence deformation fails physicality or changes the separator held fixed;
10. a singleton enriched quotient in one of the first four D5 targets, contrary to the proved continuum-strength base image and no-merger theorem. A singleton in an explicit-breaking, spontaneous, or remnant target is not a falsifier while its cardinality remains NotReady;
11. a physically valid $B - L$ -violating spectator with no target-bearing neutrino matching that is nevertheless retained as a positive explicit-breaking D5 lift. The correct disposition is spectator stripping and failure of the explicit-target matching record.

Each falsifier identifies a finite theorem boundary.

15.4 Final bounded hostile controls

The source-composition control takes $K(Y) = \{A, B, C\}$ and $\text{MinSupp}_\tau(Y) = \{\{A, B\}\}$. The required output is S: the route is unique and proper, even though it uses more than one source kind. The normalization control transports the marked conventional charge-two data to primitive integral charge six and rejects any direct inference of a $\mathbb{Z}_2$ remnant without the target-specified quotient. The Dirac-continuum controls reject a merely post-EWSB source presentation and a constant-Schur-complement argument that omits the exact physical pole spectrum and charged-current embedding. The matching controls then delete one independent light source operator or force an activity/applicability frontier crossing; the required dispositions are $\text{NotReady}_{\text{MatchingBasisIncomplete}}$ and $\text{NotReady}_{\text{ContinuationStratumCrossing}}$, respectively. The accepted continuum is the gauge-invariant, anomaly-neutral, one-apex exact-response family of Theorem 9.11 completed by the explicit source-formed finite-order matching right inverse of Theorem 9.12. These controls supplement the spectator test below.

15.5 Spectator-exclusion negative control

The adversarial control adjoins a finite, physically valid $B - L$ -violating sector Z whose internal field theory and violation record are coherent but whose source graph has no target-bearing neutrino-matching incidence. The required disposition is

$$\text{ActRed}_\nu(H \sqcup Z) = H, \quad \neg \text{ExpLift}(\bar{O}_Z; H).$$

A positive explicit-breaking classification would contradict the spectator-exclusion theorem. This control is independent of the thirteen source-representation sensitivity tests summarized in Appendix B.

15.6 Scope extensions

New information	Disposition	Required mathematical response
New source sort, constitutive relation, or infinitary architecture without finite presentation	SourceUpdate	Extend SOE–D5–1 and re-establish source completeness before evaluating the new fibres.

New information	Disposition	Required mathematical response
New anomaly probe, global form, line/defect category, operator-acceptance rule, or remnant semantics	TargetUpdate	Form a successor target rather than reinterpret an existing fibre.
Positive evidence for a specific $B - L$ gauge boson, breaking scalar, defect, or remnant	Empirical target extension	Add its response, uncertainty, and physical-interface records to a new target.
New D3 target	Separate D3 theorem	Compare only after the new target has been formed; the 1.3 and 1.1 results remain unchanged.
Complete joint D1–D5 source and survivor	PCR-5E input	Evaluate one common joint target rather than intersecting marginal answers.
Sectorwise complete protector and terminal continuation data	Post-D5 identity candidate	Apply SCIC activation, terminal eligibility and compatibility, and Role-Occupancy Closure.

15.7 Supplementary reproducibility material

The supplementary materials contain the target and fibre data, exact anomaly arithmetic, structural sensitivity tests, source and theorem provenance, and file-integrity digests. They establish the identity and internal consistency of the recorded results. They do not prove semantic source completeness, physical existence, target-image equivalence, or carrier identity; those remain mathematical claims of the paper.

A same-scope counterexample must preserve the target, source comparison class, simultaneous physical-apex condition, retained oscillation base, lawful quotient, and target tags. It must identify a failed implication, name the affected theorem, and provide an explicit construction in the same comparison class. Relabeling a target, preferring a familiar mechanism, or adding an undeclared record changes the question rather than defeating the result.

16 Conclusion

PCR-5D5 completes the marginal symmetry and charge extension that precedes the joint neutrino target. The result is not a single verdict about “ $B - L$.” It is an exact classification over seven non-equivalent targets:

$$\mathcal{F}_{D5} = \{T_{\text{acc}}^{(4)}, T_{\text{bg}, \text{min}}, T_{\text{loc}, 3N}, T_{\text{glob}}, T_{\text{exp}}[\mathcal{O}], T_{\text{sp}}[q_{\Phi}], T_{\text{rem}}[\Gamma, q_{\Phi}]\}.$$

The first four targets have nonempty proper-continuum images in the oscillation carrier atlas. Their respective separators are bounded operator invariance, the minimal background-anomaly packet, unbroken local three-singlet covariance with complete anomaly cancella-

tion, and continuation-stable exact global protection. The local/exact-global accepted-side continuum is carried by an exact physical-response vectorlike Dirac family whose complete licensed low-energy RG/EFT continuation is fixed source-first; the only standing-visible family parameter is the physical heavy threshold retained by the oriented matching channel. Explicit breaking is indexed by target-bearing source activity and neutrino matching. Spontaneous breaking is governed by the physical vacuum together with the ideal generated by the breaking charges. Remnant standing is governed by the faithful stabilizer quotient, charge-lattice congruence, and the activated global packet. None of these conclusions can be transferred to another target by relabeling.

The explicit-breaking theorem enforces the target's matching requirement. A physically valid $B - L$ -violating sector with no target-bearing neutrino-matching incidence is a PCR-3 spectator and does not furnish a D5 lift. The exact image is therefore

$$\text{im } \rho_{\text{exp}[\mathcal{O}]} = Q_{\text{ExpLift}(\mathcal{O})},$$

where ExpLift includes source support, neutrino matching, continuation, simultaneous physicality, and retained-oscillation transparency. A universal neutrino-matched transparent theorem would imply a full base image, but that universal quantifier is not established. Consequently,

$$\text{Status}(\mathcal{O} \in \mathcal{O}_{\text{D5}}^{\text{tr}}) = \text{NotReady}$$

unless a later source and physical theorem discharges it. Intrinsic target-bearing examples remain nonvacuous: the Weinberg source in q_W lies on the actual neutrino mass/matching architecture and belongs to its explicit-breaking image. Different nonzero source-activity magnitudes are not promoted into distinct standing classes without an independently admitted evaluator.

Cardinality authority is target-specific. The first four enriched targets are non-singleton because their base images contain continuum many distinct SCIC classes and D5 restriction does not merge them. The explicit, spontaneous, and remnant targets have exact image formulas, but no uniform singleton exclusion is asserted without a separate cardinality theorem. The unresolved universal explicit-breaking attachment is therefore retained as a precise NotReady boundary rather than replaced by a disconnected spectator construction.

The carrier/spectral question receives a definite but bounded answer. Exact continuous or exact residual covariance of the complete target-relevant mass response and retained spectral-incidence structure forces charge or residual character to be constant on each connected component. This conditional conclusion is $\text{ConnectedSpectralChargeUniformity}$: a symmetry-bearer theorem compatible with a common continuation-bearing carrier, but not a proof that the component is one terminal carrier. The bounded accidental $d \leq 4$ target does not inherit $\text{ConnectedSpectralChargeUniformity}$ when the complete mass response violates $B - L$. Under the exact-covariance hypothesis, mass-spectral multiplicity alone does not divide a connected component into several symmetry-distinguished terminal carriers. Terminal promotion remains outside the marginal target.

The D3 comparison closes only at the target-indexed level. Accidental conservation and exact-global protection are marginally independent of the D3-1.3 active-closure separator. The minimal-background, local-3N, explicit-breaking, spontaneous, and remnant comparisons remain NotReady because no pair of deformations has been proved to preserve their complete target packets in both directions. No common generator follows from image alignment, shared source labels, or an unresolved comparison. A common or emergent higher-arity relation remains a question for the simultaneous PCR-5E target.

The target identities, source and physical categories, lift fibres, base images, target-specific

cardinalities, separators, bearer structures, source-composition profiles, failure boundaries, spectator exclusion, and D3 comparison together define the marginal object $\mathfrak{H}_{D5}$. PCR-5E can now place that object on one common source with the other phenomenon targets. Only after that simultaneous calculation can a post-D5 identity theorem ask whether the neutrino is positively forced to be one terminal carrier with several standing spectral sectors.

Status(PCR– 5D5) = ExactTargetIndexedMixedClosure

TerminalPromotion_{D5} = NotActivated,
 CardinalityAuthority_{D5} = TargetIndexed,
 D3D5Status(1.3, $T_{acc}^{(4)}$) = IndependentCoordinate,
 D3D5Status(1.3, T_{glob}) = IndependentCoordinate,
 D3D5Status(1.3, τ) = NotReady for the remaining D5 targets,
 Status($\mathfrak{H}_{D5}$) = PCR5EReady.

A Notation and status conventions

Symbol	Meaning
T_{osc}	Fixed neutrino-oscillation target.
Q_{osc}^{SCIC}	Exact PCR-5B standing-carrier quotient of the oscillation target.
$\mathcal{N}_{osc}$	Concrete standing-core groupoid whose connected components form Q_{osc}^{SCIC} .
H_a	B15-framed PCR-3/PCR-5B source datum indexed by $a = (F, \sigma)$.
$\mathcal{F}_{D5}$	Seven-member PCR-5D5 target family.
$\mathbb{T}_\tau$	Complete target object for D5 target tag τ .
SOE–D5–1	Conservative D5 extension of the B15-framed PCR-3 source theory.
U_{D5}	Exact source reduct deleting D5-only cells.
I_{D5}	Canonical inclusion adjoining the target-typed null D5 packet to a B15 source object.
$\mathcal{G}_{\tau,a}^{src}$	Framewise source-coherent D5 presentation groupoid.
$W_{H,S}$	Simultaneous D5 physical realization of base object W_H and D5 source point S .
$\mathcal{V}_{D5}^{tr}(\tau)$	In-situ retained-oscillation transparent physical locus.
R_τ	Concrete standing-core restriction functor.
ρ_τ	Descended map from the enriched D5 quotient to Q_{osc}^{SCIC} .
$\mathcal{C}_\tau(q)$	Connected homotopy lift groupoid over base class q .
Q_P	Base subatlas whose connected classes admit a representative satisfying invariant predicate P .
$Acc_{\leq 4}$	Every source-active $d \leq 4$ operator preserves $B - L$.
MinBG	Minimal chiral inventory and complete background anomaly packet.

Symbol	Meaning
Loc_{3N}	Faithful unbroken local $U(1)_{B-L}$, one active ν_R per family, covariant support, and zero local polynomial anomaly tuple.
GlobProt	Faithful exact global action with a continuation-stable protector.
$\mathcal{O}_{D5}^{\text{tr}}$	Derived universal class of operator types admitting a target-bearing neutrino-matched, oscillation-transparent lift over every base class; nonemptiness status NotReady.
$\mathcal{O}_\star$	Physically valid sterile-singlet Majorana spectator used as the negative control for spectator stripping when no target-bearing neutrino matching is present.
$\text{ExpLift}(\mathcal{O})$	Physical source-active explicit-breaking lift of operator or process class $\mathcal{O}$.
$\text{Cov}_\Phi(q_\Phi)$	Physical spontaneous lift whose violation ideal is contained in the ideal generated by q_Φ .
$\text{RemFaith}(\Gamma, q_\Phi)$	Physical spontaneous lift with faithful stabilizer quotient Γ and quotient-invariant retained operators.
$\mathfrak{B}_{D5}(Y)$	Complete D5 bearer-incidence hypergraph of enriched point Y .
$\mathcal{C}_{\text{spec}}(Y)$	Retained spectral-incidence graph of a target-complete neutrino carrier.
ConnectedSpectral ChargeUniformity	Connected spectral charge uniformity; compatible with, but not proof of, one terminal carrier.
SectorwiseSpectral SymmetryStanding	Sectorwise spectral symmetry standing.
SectorwiseComplete ProtectorSignature	Sectorwise independently complete protector signature.
RelationalHigherArity SymmetryStanding	Relational or higher-arity symmetry standing.
BearerNotReady	Bearer status unresolved because required data are incomplete.
S/R/P/Y	Single-route, redundant, partially synergistic, and synergistic no-repair source-composition classes, assigned by the precedence rule in Section 12.
$\mathfrak{H}_{D5}$	Complete target-tagged D5 marginal object used in PCR-5E.

A.1 Source-composition and charge-normalization conventions

For an accepted source point Y , the finite antichain $\text{MinSupp}_\tau(Y)$ consists of the inclusion-minimal no-repair sufficient source subsets. The reported source-composition class is assigned by precedence: Y when the full active set is uniquely minimal; S when there is one unique proper minimal route; P when there are at least two minimal routes and every one uses more than one source kind; and R when there are at least two minimal routes and at least one is a singleton. This makes the four outputs exhaustive and pairwise disjoint.

All perturbative anomaly sums use left-handed Weyl fields. Physical right-handed fields are represented by their left-handed conjugates. Two charge coordinates are used and never mixed inside one arithmetic calculation.

In the conventional rational normalization,

$$\begin{aligned} q_{B-L}^{\text{conv}}(Q) &= \frac{1}{3}, & q_{B-L}^{\text{conv}}(u^c) &= q_{B-L}^{\text{conv}}(d^c) = -\frac{1}{3}, \\ q_{B-L}^{\text{conv}}(L) &= -1, & q_{B-L}^{\text{conv}}(e^c) &= q_{B-L}^{\text{conv}}(\nu^c) = +1. \end{aligned}$$

The displayed label $\Delta(B - L)$ denotes total source-operator charge in this conventional coordinate unless an “int” superscript is shown. For divisibility, Smith-normal-form, and remnant arithmetic, let D_Λ be the least common denominator of the target-active faithful charge lattice and use the primitive integral transport

$$q^{\text{int}} = D_\Lambda q^{\text{conv}}, \quad \delta_O^{\text{int}} = D_\Lambda \Delta(B - L)_O^{\text{conv}}.$$

For the marked Standard-Model lattice, $D_\Lambda = 3$: quark-doublet charge is 1, lepton-doublet charge is -3 , a conventional $|\Delta(B - L)| = 2$ source has $|\delta_O^{\text{int}}| = 6$, and a conventional charge-two breaking scalar has $|q_\Phi^{\text{int}}| = 6$. Its raw local stabilizer is $\mathbb{Z}_6$; a faithful $\mathbb{Z}_2$ remnant arises only when the declared global-form packet additionally quotients an ineffective order-three subgroup. Only vanishing charge or the charge class in the actual declared faithful quotient is standing-bearing. Reversing every charge is lawful when the complete lattice, action, coupling, ineffective-kernel, and global-quotient data transform with it.

A.2 Fibre geometry

A fibre is *stable* when it has one connected component after lawful quotienting; *finite* or *countable* when its component set has that cardinality; *continuous* when it contains a nondegenerate continuum of standing-distinct components; *functional* when function-space moduli survive; *categorical* when nontrivial groupoid or higher-overlap data remain; and *mixed* when more than one form is present. A typed NotReady is not an empty fibre.

B Source representation and adversarial controls

B.1 The source-representation result

The D5 fibre analysis relies on the previously established source-representation theorem

$$\text{Status}(\text{SLR}-\text{D5}-1) = \text{SourceLandingCertified}.$$

The theorem establishes source representation, source completeness, and conservativity over the B15 base. It does not decide anomaly cancellation, physical survival, fibre nonemptiness, base images, bearer status, marginal cardinality, or the D3 comparison.

B.2 Established source-level properties

Its source-level conclusions are:

1. Every record in the seven-member D5 target family lands in a D5 source sort, operation, relation, or typed null terminal.
2. Every finite same-target symmetry and charge architecture in the declared comparison class has a source-coherent normal form.
3. Every generated coherent normal form has a semantic source interpretation.
4. The reduct U_{D5} returns the exact PCR-3/B15 source object and preserves higher-arity in-

cidence.

5. The source-conservative locus preserves the complete retained oscillation trace before any physical deletion.
6. Local, global, accidental, explicit, spontaneous, and remnant records remain separately typed.
7. The bearer grammar covers connected spectral charge uniformity, sectorwise spectral standing, complete sectorwise protector standing, relational or higher-arity standing, and unresolved cases without terminal promotion.
8. The source theory is sensitive to each of the thirteen structural perturbations listed below and assigns the required failure or update status.

B.3 Adversarial source tests

ID	Perturbation	Required response
M01	Delete the cubic and mixed-gravitational anomaly components while retaining $T_{bg,min}$.	Reject the incomplete anomaly packet.
M02	Promote local three- ν_R cancellation to globally complete gauged $B - L$.	Reject the absent global-completion data.
M03	Treat accidental $B - L$ as exact-global protection.	Reject the target coercion.
M04	Set a $\Delta L = 2$ coefficient to zero and declare exact symmetry.	Reject a parameter value as a protector.
M05	Assign a remnant label without a breaking carrier, charge lattice, and stabilizer.	Reject the missing derivation data.
M06	Combine anomaly, vacuum, and matching records from different physical points.	Reject the missing common realization.
M07	Use D2 non-observation to select exact Dirac character.	Reject a marginal null result as a protector.
M08	Use the D3 proper image as the D5 base image.	Reject cross-target membership.
M09	Convert a family label or mass projector into a terminal carrier by definition.	Reject terminal promotion.
M10	Infer three right-handed neutrinos solely from $N_g = 3$.	Reject generation-to-occupancy inference.
M11	Remove an active global-form, line, or defect record while retaining the target.	Reject the incomplete global packet.
M12	Merge target fibres because their base images agree.	Reject image-based target identity.
M13	Declare a D3–D5 common generator from equal images without factorization.	Reject the missing common invariant.

The supplementary concordance records the required disposition in all thirteen cases. These controls demonstrate sensitivity to the declared distinctions; they do not replace the semantic source-representation proof.

C Marked source witnesses and continuum calibrations

This appendix supplies explicit positive and negative witnesses used in the properness and cardinality arguments. They are not a replacement for the complete source atlas; they certify that the exact subatlases are nonempty and proper and, where stated, contain continuum families.

C.1 Law-only Weinberg witness q_W

Let the active source be the Standard Model field inventory plus a source-active dimension-five Weinberg operator

$$\mathcal{O}_5^{\alpha\beta} = (L_\alpha^T C L_\beta) H H$$

with nonzero coefficient. Its conventional rational $B - L$ charge is -2 , corresponding to primitive integral charge -6 in the marked faithful lattice. It has no active $d \leq 4$ $B - L$ -violating source term. Therefore

$$q_W \in Q_{\text{Acc}_{\leq 4}} \cap Q_{\text{MinBG}}, \quad q_W \notin Q_{\text{Loc}_{3N}} \cup Q_{\text{GlobProt}}.$$

It belongs intrinsically to the explicit-breaking image for $\mathcal{O}_5$ and can be covariantized by a breaking carrier whose primitive integral charge divides 6.

C.1.1 A continuum LAW calibration

Write the low-energy source coefficient as $\kappa = C_5/\Lambda$. Fix a nonzero oscillation-compatible κ_* and choose

$$\Lambda_\lambda = \lambda \Lambda_0, \quad C_{5,\lambda} = \lambda C_{5,0}, \quad \lambda \in (1, 2).$$

Then $C_{5,\lambda}/\Lambda_\lambda = \kappa_*$, so the low-energy oscillation response is fixed, while the oriented matching threshold Λ_λ changes. The PCR-5B standing packet retains matching orientation and threshold, and lawful LAW redescription does not scale the physical threshold. Hence

$$\lambda \neq \mu \implies [q_{W,\lambda}] \neq [q_{W,\mu}] \text{ in } Q_{\text{osc}}^{\text{SCIC}}.$$

The family gives a continuum inside $Q_{\text{Acc}_{\leq 4}}$ and Q_{MinBG} .

C.2 Renormalizable Majorana active-singlet witness q_{3N}

Fix a full-rank symmetric low-energy coefficient $\kappa_* = U D U^T$ with $D > 0$, a positive nonsingular singlet matrix M_0 , and

$$Y_0 = U D^{1/2} M_0^{1/2}.$$

For $\lambda \in (1, 2)$ set

$$M_\lambda = \lambda M_0, \quad Y_\lambda = \sqrt{\lambda} Y_0.$$

Then

$$Y_\lambda M_\lambda^{-1} Y_\lambda^T = \kappa_*$$

while the physical threshold spectrum changes. This is the PCR-5B active-singlet continuum. Because $M_\lambda \neq 0$ is a dimension-three source with conventional $|\Delta(B - L)| = 2$ and primitive integral violation $|\delta_{\mathcal{O}}^{\text{int}}| = 6$, and the chiral inventory contains active singlets,

$$q_{3N,\lambda} \notin Q_{\text{Acc}_{\leq 4}} \cup Q_{\text{MinBG}} \cup Q_{\text{Loc}_{3N}} \cup Q_{\text{GlobProt}}.$$

It belongs to the intrinsic explicit-breaking image for the singlet Majorana operator and to spontaneous/remnant branches with primitive integral breaking charge 6 when their full physical packets exist.

C.3 Exact-Dirac three-singlet witness q_D

Take three active right-handed neutrinos with a faithful exact $B-L$ action and a continuation-stable Protect relation. Source covariance excludes the Majorana source and every other active $\Delta(B-L) \neq 0$ incidence throughout the licensed continuation grammar; the vanishing of M_R is therefore a consequence of the protected source architecture, not a tuned parameter value. Given an oscillation-compatible Dirac mass matrix M_D , define

$$Y_\nu = \frac{\sqrt{2}}{\nu} M_D.$$

The source operators $\bar{L}\widetilde{H}\nu_R$ preserve $B-L$, and one ν_R per family cancels the cubic and mixed-gravitational deficits. With the faithful action and protector in force, this gives

$$q_D \in Q_{\text{Acc}_{\leq 4}} \cap Q_{\text{Loc}_{3N}} \cap Q_{\text{GlobProt}}, \quad q_D \notin Q_{\text{MinBG}}.$$

The last exclusion is target-relative: $T_{\text{bg,min}}$ fixes the minimal chiral inventory, not an anomaly-free extension. The local target requires one source-active ν_R per family but does not prohibit additional anomaly-neutral vectorlike support. Such support is admissible only when it preserves the faithful local action, covariant source equations, complete anomaly packet, matching, and retained oscillation standing.

C.3.1 Detailed physical-response record for the vectorlike Dirac threshold continuum

The accepted-side continuum for the local and exact-global targets is the source family of Theorem 9.11. Its defining feature is exact control of the complete physical spectrum and boundary response, not merely a constant Schur complement.

Fix

$$M_D^* = U\Sigma V^\dagger, \quad \Sigma = \text{diag}(\sigma_1, \sigma_2, \sigma_3), \quad \sigma_i > 0,$$

and choose $\Lambda > E_{\text{osc}}^{\text{max}}$ and

$$0 < s < \min\left\{\frac{1}{4}, \frac{\nu}{8\sqrt{2}\Lambda}\right\}, \quad c = \sqrt{1-s^2},$$

so that the fixed event-complete light packet with charged-current matrix cU , light singular values Σ , and its null/no-event record is target-admissible. For $\lambda \in (1, 2)$ set $H_\lambda = \lambda\Lambda$ and

$$\begin{aligned} Y_\nu &= \frac{\sqrt{2}}{\nu} cU\Sigma V^\dagger, & Y_{LN,\lambda} &= -\frac{\sqrt{2}}{\nu} sH_\lambda U, \\ \mu &= s\Sigma V^\dagger, & M_{N,\lambda} &= cH_\lambda \mathbf{1}. \end{aligned}$$

For equal-rank Standard-Model-singlet multiplets N_L, N_R of hypercharge zero and conventional $B-L = -1$, use

$$-\bar{L}\widetilde{H}Y_\nu\nu_R - \bar{L}\widetilde{H}Y_{LN,\lambda}N_R - \overline{N_L}\mu\nu_R - \overline{N_L}M_{N,\lambda}N_R + \text{h.c.}$$

These operators are invariant under $SU(2)_L \times U(1)_Y \times U(1)_{B-L}$. The vectorlike pair contributes opposite left-handed $B-L$ charges, so its linear and cubic anomalies cancel and the

original three chiral ν_R remain the local anomaly-cancelling packet.

After electroweak symmetry breaking,

$$\mathbb{M}_\lambda = \begin{pmatrix} cU\Sigma V^\dagger & -sH_\lambda U \\ s\Sigma V^\dagger & cH_\lambda \mathbf{1} \end{pmatrix} = \mathcal{U}_L \begin{pmatrix} \Sigma & 0 \\ 0 & H_\lambda \mathbf{1} \end{pmatrix} \mathcal{U}_R^\dagger,$$

with

$$\mathcal{U}_L = \begin{pmatrix} U & 0 \\ 0 & \mathbf{1} \end{pmatrix} \begin{pmatrix} c\mathbf{1} & -s\mathbf{1} \\ s\mathbf{1} & c\mathbf{1} \end{pmatrix}, \quad \mathcal{U}_R = \begin{pmatrix} V & 0 \\ 0 & \mathbf{1} \end{pmatrix}.$$

Hence the exact light singular values are σ_i , the exact heavy singular values are H_λ , and the left light/heavy charged-current matrices are the λ -independent blocks cU and $-sU$. The kinetic terms remain canonical because the diagonalization is unitary. Since $H_\lambda > E_{\text{osc}}^{\text{max}}$ throughout the interval, the heavy direct-production status is fixed and the complete target-energy light response is independent of λ . The family therefore preserves the exact physical oscillation evaluator while changing only the retained physical threshold and its oriented matching data.

The nonzero active–vectorlike incidences and the nonzero heavy charged-current projection prevent spectator stripping. No lawful source, basis, gauge, charge-normalization, or chart arrow changes the physical heavy singular-value spectrum. Therefore

$$\lambda \neq \mu \implies [\mathbf{x}_\lambda]_{\text{SCIC}} \neq [\mathbf{x}_\mu]_{\text{SCIC}}, \quad |Q_{\text{Loc}_{3N}} \cap Q_{\text{GlobProt}}| \geq \mathfrak{c}.$$

A raw Schur complement is not used as a substitute for this physical calculation; exact pole masses, canonical normalization, charged-current embedding, and source-scale applicability are read from the displayed unitary factorization. The complete licensed continuation is supplied separately by Theorem 9.12.

C.3.2 Standing-preserving finite-order RG/EFT matching record

Fix the finite B15 EFT envelope through order K_{EFT} , a gauge- and exact- $B-L$ -covariant renormalization prescription, and a threshold-matching convention. After quotienting by integration by parts, equations of motion, Ward identities, and lawful field redefinitions, choose the finite independent basis $\mathcal{O}_{a,n}$ dual to the continuous primitive coordinates $r_{a,n}$. The finite matching equations have the triangular form

$$r_{a,n}^{\text{match}}(\lambda) = p_{a,n}^{\text{src}}(\lambda) + \Delta r_{a,n}(\lambda; p_{<n}^{\text{src}}(\lambda)).$$

Theorem 9.12 forms, before target evaluation, the explicit recursive right inverse

$$p_{a,n}^{\text{src}}(\lambda) = r_{a,n}^* - \Delta r_{a,n}(\lambda; p_{<n}^{\text{src}}(\lambda)),$$

so every matched coefficient equals its benchmark value. The ordered source-to-matched matrix is triangular with unit diagonal. Equivalently, after the finite recursion is assembled,

$$C^{\text{match}}(\lambda) = C^{\text{src}}(\lambda) + \Delta C(\lambda) = C^*.$$

No Jacobian regularity, local-surjectivity hypothesis, or implicit-function theorem is used. A source-formed recursive heavy on-shell subtraction simultaneously fixes

$$\text{sv}_{\text{heavy}}(\widehat{\mathbf{x}}_\lambda) = H_\lambda \mathbf{1}, \quad H_\lambda = \lambda \Lambda.$$

The retained data are separated into $\mathcal{R}_{\text{light}}^{\text{cont}}$ and $\mathcal{S}_{\text{light}}^{\text{disc}}$. The explicit right inverse fixes the for-

mer exactly. On a nondegenerate threshold-free interval I_{match} , the source activity pattern and every applicability, response-mode, and failure status are constant, so the latter is fixed without treating discrete statuses as analytic variables. The complete continuation factors as

$$\text{Cont}_{\text{B15}}(\widehat{X}_\lambda) = (\text{Cont}_{\text{light}}^*, \text{Match}_{H_\lambda}, \text{Cont}_{\text{heavy}, \lambda}).$$

All source coefficients and counterterms are fixed before target evaluation. If the independent basis is incomplete the disposition is $\text{NotReady}_{\text{MatchingBasisIncomplete}}$; if the interval crosses a discrete frontier it is $\text{NotReady}_{\text{ContinuationStratumCrossing}}$. This explicit finite-order right inverse is the continuation-level reason that the threshold continuum lies in $Q_{\text{Loc}_{3N}} \cap Q_{\text{GlobProt}}$.

C.4 Target-bearing explicit breaking and the spectator control

The explicit-breaking target has both positive intrinsic witnesses and a decisive spectator negative control.

Intrinsic target-bearing positive witness. The marked law-only class q_W contains the source-active Weinberg operator

$$\mathcal{O}_5^{\alpha\beta} = (L_\alpha^T C L_\beta) H H, \quad |\Delta(B - L)_{\mathcal{O}_5}^{\text{conv}}| = 2, \quad |\delta_{\mathcal{O}_5}^{\text{int}}| = 6,$$

as part of its actual neutrino mass/matching architecture. The operator therefore lies on a current target-relative carrier path rather than in an auxiliary decoupled sector, and

$$q_W \in Q_{\text{ExpLift}(\mathcal{O}_5)}.$$

This establishes nonvacuity for that intrinsic explicit target. It does not establish a universal full-base attachment theorem.

Spectator negative control. Let N_s be an auxiliary left-handed Standard-Model singlet of conventional rational $B - L$ charge +1 (primitive integral charge +3 on the marked faithful lattice) with Majorana process

$$\mathcal{O}_* = \frac{1}{2} N_s^T C N_s + \text{h.c.}, \quad |\Delta(B - L)_{\mathcal{O}_*}^{\text{conv}}| = 2, \quad |\delta_{\mathcal{O}_*}^{\text{int}}| = 6.$$

Its local field theory may be perfectly physical. If it is adjoined to a B15 source H with no source-active D5 neutrino-matching incidence, however, PCR-3 activity reduction makes it a target-inert spectator:

$$\text{ActRed}_\nu(H \sqcup N_s) = H, \quad \neg \text{ExpLift}(\mathcal{O}_*; H).$$

A common charge label, common physical apex, or nonzero violation is insufficient to override this result. The example is therefore an adversarial negative control rather than a positive member of $\mathcal{O}_{\text{D5}}^{\text{tr}}$.

Universal matched-transparent status. The derived class $\mathcal{O}_{\text{D5}}^{\text{tr}}$ requires an already admitted operator or process to have a target-bearing neutrino-matching incidence and zero projection under every retained oscillation evaluator and continuation over every base class. The stated D5 source and physical hypotheses do not prove such a universal family nonempty,

so

$$\text{Status}(\mathcal{O}_{D5}^{\text{tr}} \neq \emptyset) = \text{NotReady}.$$

C.5 Stability of the continuum witnesses

Lemma C.1 (Packet-rigid exclusion witnesses). *The continuum witnesses used to prove properness are base-class witnesses, not merely failed source presentations. In the LAW scaling family, the fixed activity, law/new-local deletion, and oriented matching channels retain the active law support and its matching threshold. In the active-singlet Majorana scaling family, the fixed activity and matching channels retain the active singlet graph, nonzero Majorana threshold object, and matching orientation. Therefore any B15 presentation of the same connected SCIC class carries an isomorphic retained standing core with the same relevant active-support and matching data. A presentation that removes those incidences or replaces their matching orientation lies in a different base class rather than repairing the D5 target within the fibre.*

Proof. PCR-5B identifies base classes by the complete seven-channel packet and forms overlaps over the concrete standing-core groupoid. The channels C^{act} , M^{match} , and the fixed no-repair deletion record are objectwise invariant under every overlap arrow. The two scaling paths change physical thresholds but preserve their source presentation, active incidence type, deletion type, and matching orientation. Hence an overlap representative must preserve those data. Deleting the law support, nulling the Majorana incidence, or substituting a Dirac matching arrow changes at least one fixed channel, so packet completeness forbids identification with the original connected class. The target failures used in Section 9 consequently hold over every frame of the displayed continuum classes. $\square$

C.6 Mixed marked witnesses

The PCR-5A/PCR-5B marked mixed points have the following source-semantic dispositions.

Class	Deletion pair	Active conventional $ \Delta(B - L) = 2$ support	D5 consequence
q_W	(0, 1)	Weinberg law	Accidental/minimal-background pass; local/global fail; explicit $\mathcal{O}_5$ pass.
q_{3N}	(1, 0)	Singlet Majorana source	Accidental/minimal-background/local/global fail; explicit M_R pass.
q_{red}	(1, 1)	Redundant Weinberg and singlet-Majorana routes	Explicit breaking can be redundant; unbroken local/global fail.
q_{syn}	(0, 0)	Rank-split jointly necessary conventional $ \Delta(B - L) = 2$ routes (primitive integral $ \delta = 6$)	Explicit breaking is synergistic on the marked source; unbroken local/global fail.

Class	Deletion pair	Active conventional $ \Delta(B - L) = 2$ support	D5 consequence
q_D	direct local	none	Accidental/local/global pass; the intrinsic explicit target with conventional $ \Delta(B - L) = 2$ (primitive integral $ \delta = 6$) fails.

C.7 Spontaneous and remnant calibrations

The calibration uses one normalization throughout. Conventional rational charges are transported by $q^{\text{int}} = 3q^{\text{conv}}$ on the marked faithful lattice. Thus the Weinberg and singlet-Majorana violations conventionally labelled $|\Delta(B - L)| = 2$ have $|\delta_O^{\text{int}}| = 6$, and a scalar conventionally labelled $|B - L| = 2$ has $|q_\Phi^{\text{int}}| = 6$. Inserting Φ or $\Phi^\dagger$ therefore lifts both marked violations to invariant pre-breaking operators at the arithmetic level.

A single $|q_\Phi^{\text{int}}| = 6$ scalar leaves the raw local stabilizer $\mathbb{Z}_6$. The faithful remnant remains $\mathbb{Z}_6$ unless the declared global-form packet identifies an ineffective order-three subgroup, in which case the faithful quotient is $\mathbb{Z}_2$. Since $6 \equiv 0 \pmod{6}$ and also maps to the trivial class in the quotient $\mathbb{Z}_2$, both targets permit Dirac and Majorana-compatible mass support. Neither remnant is therefore an exact-Dirac selector.

For a faithful one-scalar remnant $\mathbb{Z}_n$, the marked Majorana/Weinberg operators survive exactly when $n \mid 6$ and fail when $n \nmid 6$. This produces a target-relative exclusion from the actual faithful quotient; the inequality $n > 2$ alone has no invariant selection meaning, and multi-scalar or different global-form completions require their own target calculation.

C.8 Continuum-strength summary

The LAW scaling family supplies continuum accepted witnesses for $T_{\text{acc}}^{(4)}$ and $T_{\text{bg},\text{min}}$. The exact-response $B - L$ -preserving Dirac threshold family supplies continuum accepted witnesses for $T_{\text{loc},3N}$ and T_{glob} . The active-singlet Majorana scaling family and the LAW family supply continuum complements. These explicit intervals justify the proper-continuum claims in Section 11 without counting family names or assuming generic parameter variation.

D Exact anomaly arithmetic and reproducibility data

D.1 One-family anomaly calculation

Use the left-handed Weyl spectrum

$$\begin{aligned} Q &: (\mathbf{3}, \mathbf{2})_{1/6}^{1/3}, & u^c &: (\bar{\mathbf{3}}, \mathbf{1})_{-2/3}^{-1/3}, & d^c &: (\bar{\mathbf{3}}, \mathbf{1})_{1/3}^{-1/3}, \\ L &: (\mathbf{1}, \mathbf{2})_{-1/2}^{-1}, & e^c &: (\mathbf{1}, \mathbf{1})_1^{+1}, & \nu^c &: (\mathbf{1}, \mathbf{1})_0^{+1}. \end{aligned}$$

The subscript is hypercharge and the superscript is $B - L$ charge. Physical right-handed fields are represented by their left-handed conjugates, and the minimal spectrum omits ν^c .

The one-family coefficients are

$$\begin{aligned}
A_{SU(3)^2-(B-L)} &= 2T(\mathbf{3})\frac{1}{3} + T(\bar{\mathbf{3}})\left(-\frac{1}{3}\right) + T(\bar{\mathbf{3}})\left(-\frac{1}{3}\right) = 0, \\
A_{SU(2)^2-(B-L)} &= 3T(\mathbf{2})\frac{1}{3} + T(\mathbf{2})(-1) = 0, \\
A_{Y^2-(B-L)} &= 6\left(\frac{1}{6}\right)^2\frac{1}{3} + 3\left(-\frac{2}{3}\right)^2\left(-\frac{1}{3}\right) + 3\left(\frac{1}{3}\right)^2\left(-\frac{1}{3}\right) \\
&\quad + 2\left(-\frac{1}{2}\right)^2(-1) + 1^2(+1) = 0, \\
A_{Y-(B-L)^2} &= 6\left(\frac{1}{6}\right)\left(\frac{1}{3}\right)^2 + 3\left(-\frac{2}{3}\right)\left(-\frac{1}{3}\right)^2 + 3\left(\frac{1}{3}\right)\left(-\frac{1}{3}\right)^2 \\
&\quad + 2\left(-\frac{1}{2}\right)(-1)^2 + 1(+1)^2 = 0, \\
A_{\text{grav}^2-(B-L)}^{\min} &= 6\left(\frac{1}{3}\right) + 3\left(-\frac{1}{3}\right) + 3\left(-\frac{1}{3}\right) + 2(-1) + (+1) = -1, \\
A_{(B-L)^3}^{\min} &= 6\left(\frac{1}{3}\right)^3 + 3\left(-\frac{1}{3}\right)^3 + 3\left(-\frac{1}{3}\right)^3 + 2(-1)^3 + (+1)^3 = -1.
\end{aligned}$$

Adding ν^c contributes +1 to the last two rows and zero to the first four. For three families,

$$\mathfrak{A}_{\min} = (0, 0, 0, 0, -3, -3), \quad \mathfrak{A}_{3\nu_R} = (0, 0, 0, 0, 0, 0).$$

The exact rational calculation is reproduced in `outputs/anomaly_calculation.json`.

D.2 Structured target and fibre atlas

The supplementary concordance `PCR5D5_Target_Fibre_Atlas_v1.2.5.json` gives, for each target, its fingerprint, exact source predicate, base-image expression, required and forbidden records, distinguished witnesses, fibre geometry, target-specific cardinality status, minimal separator, bearer and terminal-promotion status, target-indexed D3 comparison, and the conditions under which each conclusion changes. It is a structured concordance of the theorem outputs and does not substitute for their proofs.

D.3 Theorem concordance

`PCR5D5_Theorem_Ledger_v1.2.5.csv` relates D5-01 through D5-12 to their dependencies, proof dependencies, scientific conclusions, counterexample criteria, and later use. The consistency analysis checks uniqueness of theorem identifiers, totality of target tags, and agreement with the target atlas.

D.4 Marked-class dispositions

`PCR5D5_Marked_Class_Dispositions_v1.2.5.csv` records the target-wise status of q_W , q_{3N} , q_{red} , q_{syn} , and q_D . The companion `D5_Dirac_Threshold_Continuum_v1.2.5.json` records the finite source terms, charge/anomaly checks, exact unitary physical factorization, light/heavy pole spectrum, canonical charged-current response, applicability status, physical-apex clauses, and threshold-separation data. The separate `D5_RG_EFT_Transport_v1.2.5.json` records the quotient-reduced independent matching basis, explicit source-formed finite-order right inverse, continuous/discrete packet separation, continuation decomposition, and threshold-only standing variation of the accepted local/global continuum witness.

The numerical consistency record `finite_order_matching_right_inverse_check_v1.2.5.json` verifies representative affine matching identities and transport invertibility; it is a reproducibility aid, not a proof of operator-basis completeness. These files are concordances of the proofs, not substitutes for them. The marked-class values lie in

{Pass, Fail, Conditional, N/A, NotReady}

and each target-class entry carries a separator code. No entry is permitted to infer terminal-carrier identity.

D.5 Scope of the consistency checks

The supplementary consistency check verifies internal document identity, the seven target tags and twelve theorem identifiers, exact anomaly arithmetic, cross-fibre no-backflow rules, the terminal-promotion codomain, D3 target-version tagging, target-indexed comparison statuses, preservation of explicit NotReady frontiers, the independent definition of the neutrino-matched transparency predicate, the NotReady status of universal $\mathcal{O}_{D5}^{\text{tr}}$ nonemptiness, spectator stripping for an unmatched physical breaker, the target-bearing intrinsic positive witness, the optional downstream Julia–Halmos representation boundary, the rejection of Schur-complement-only response matching, the exact physical-response continuum record, the source-formed finite-order RG/EFT matching right-inverse record and hostile controls, and SHA-256 file identity. These checks support reproducibility. Semantic completeness, physical existence, and mathematical necessity remain theorem burdens.

E Dependency provenance and imported theorem roles

E.1 Controlling sources

The controlling scientific sources and their exact roles are:

Source	Exact imported role	Use boundary
PCR-3, <i>Source-First Exhaustion of the Physical Carrier Universe</i>	Finite source ontology, activity reduction, normal forms, finite hybrid closure.	Source membership is not D5 survival.
PCR-4, <i>Simultaneous Physical Constraint Collapse</i>	One-point semantic physicality and common-cone constraint.	Marginal records may not be spliced.
PCR-5B, <i>Exact Standing-Carrier Stratification</i>	Exact B15 groupoid atlas, packet, overlaps, complement, continuum witness.	Base identity is held fixed.
PCR-5C, <i>Functorial Target Extension</i>	Restriction, no merger, connected fibres, codescent, source composition, failure provenance.	Introduces no D5-specific physical premise.
D1–D4 carrier/spectral synthesis	Target-qualified standing-coordinate comparison and carrier/spectral firewall.	Hypothesis is not a D5 selector.

Source	Exact imported role	Use boundary
SYM-7, <i>Accidental Symmetry by Operator Dimension</i>	Neutrino target separation; $B - L$ anomaly and breaking/remnant source theorems.	Local polynomial result is not global completion.
SYM-8, <i>Electroweak Vacuum Fixation and Residual Spectrum Rigidity</i>	Vacuum orbit, stabilizer, residual action, no representative selection.	A vev label does not establish a branch.
SYM-9, <i>Global Form and Generalized Symmetry</i>	Global completion packet, line/defect/global-anomaly and fixed/indexed target discipline.	No unique global form is selected.
SYM-11, <i>Standard Model Symmetry Atlas and Exhaustion Closure</i>	Source-first symmetry normal forms and target-qualified closure.	Does not precompute D5 fibres.
SM carrier/cardinality stack	Matter, charge, flavour, Higgs-bridge roles; $N_g = 3$ governance.	Generation count is not ν_R occupancy or carrier count.
Standing Carrier Identity Closure	Final same-scope carrier quotient and nonpromotion of unwitnessed distinctions.	Requires an activated complete carrier interface.
Role-Occupancy Closure	Terminal eligibility, compatibility, and unique assignment orbit.	Outside marginal D5.
PCR-5D3 successor publication v1.2.0	Target-specific active-closure separator and proper-continuum result at THUCL-D3-1.3.	Does not transport to THUCL-D3-1.1.

E.2 Dependency polarity

The imported chain is used only in the direction

source completeness $\rightarrow$ physical one-apex evaluation $\rightarrow$ retained restriction
 $\rightarrow$ fibre exhaustion $\rightarrow$ D5 target classification
 $\rightarrow$ bearer analysis and PCR-5E interface.

No downstream D5 image or desired carrier interpretation is used to alter the source grammar, target family, anomaly convention, or bearer codomain.

E.3 Source identification

For sources included in the supplementary materials, `SOURCE_HASH_LOCK.csv` records the archival filename and SHA-256 digest. Other sources are identified by title, target/version identifier, and theorem role. Use of a different source version requires either a theorem-equivalence argument or a new dependency analysis with the relevant consistency checks.

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